

CHAPTER C15

SEISMIC DESIGN REQUIREMENTS FOR NONBUILDING STRUCTURES

C15.1 GENERAL

C15.1.1 Nonbuilding Structures. Building codes traditionally have been perceived as minimum standards for the design of nonbuilding structures, and building code compliance of these structures is required by building officials in many jurisdictions. However, requirements in the industry reference documents are often at odds with building code requirements. In some cases, the industry documents need to be altered, whereas in other cases, the building codes need to be modified. Registered design professionals are not always aware of the numerous accepted documents within an industry and may not know whether the accepted documents are adequate. One of the intents of Chapter 15 of the standard is to bridge the gap between building codes and existing industry reference documents.

Differences between the ASCE/SEI 7 design approaches for buildings and industry document requirements for steel multilegged water towers (Fig. C15.1-1) are representative of this inconsistency. Historically, such towers have performed well when properly designed in accordance with American Water Works Association (AWWA) standards and industry practices. Those standards and practices differ from the ASCE/SEI 7 treatment of buildings in that tension-only rods are allowed,



FIGURE C15.1-1 Steel multilegged water tower

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upset rods are preloaded at the time of installation, and connection forces are not amplified.

Chapter 15 also provides an appropriate link so that the industry reference documents can be used with the seismic ground motions established in the standard. Some nonbuilding structures are similar to buildings and can be designed using sections of the standard directly, whereas other nonbuilding structures require special analysis unique to the particular type of nonbuilding structure.

Building structures, vehicular bridges, electrical transmission towers, hydraulic structures (e.g., dams), buried utility lines and their appurtenances, and nuclear reactors are excluded from the scope of the nonbuilding structure requirements, although industrial buildings are permitted per Chapter 11 to use the provisions in Chapter 15 for nonbuilding structures with structural systems similar to buildings, provided that specific conditions are met. The excluded structures are covered by other well-established design criteria (e.g., electrical transmission towers and vehicular bridges), are not under the jurisdiction of local building officials (e.g., nuclear reactors and dams), or require technical considerations beyond the scope of the standard (e.g., buried utility lines and their appurtenances).

C15.1.2 Design. Nonbuilding structures and building structures have much in common with respect to design intent and expected performance, but there are also important differences. Chapter 15 relies on other portions of the standard where possible and provides special notes where necessary.

There are two types of nonbuilding structures: those with structural systems similar to buildings and those with structural systems not similar to buildings. Specific requirements for these two cases appear in Sections 15.5 and 15.6.

C15.1.3 Structural Analysis Procedure Selection. Nonbuilding structures that are similar to buildings are subject to the same analysis procedure limitations as building structures. Nonbuilding structures that are not similar to buildings are subject to those limitations and are subject to procedure limitations prescribed in applicable specific reference documents.

For many nonbuilding structures supporting flexible system components, such as pipe racks (Fig. C15.1-2), the supported piping and platforms generally are not regarded as rigid enough to redistribute seismic forces to the supporting frames.

For nonbuilding structures supporting very stiff (i.e., rigid) system components, such as steam turbine generators (STGs) and heat recovery steam generators (HRSGs) (Fig. C15.1-3), the supported equipment, ductwork, and other components (depending on how they are attached to the structure) may be rigid enough to redistribute seismic forces to the supporting frames. Torsional effects may need to be considered in such situations.



FIGURE C15.1-2 Steel pipe rack

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FIGURE C15.1-3 Heat recovery steam generators

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Section 12.6 presents seismic analysis procedures for building structures based on the Seismic Design Category (SDC); the fundamental period, T ; and the presence of certain horizontal or vertical irregularities in the structural system. Where the fundamental period is greater than or equal to $3.5 T_s$ (where $T_s = S_{D1}/S_{DS}$), the use of the equivalent lateral force procedure is not permitted in SDCs D, E, and F. This requirement is based on the fact that, unlike the dominance of the first mode response in case of buildings with lower first mode period, higher vibration modes do contribute more significantly in situations when the first mode period is larger than $3.5 T_s$. For buildings that exhibit classic flexural deformation patterns (such as slender shear-wall or braced-frame systems), the second mode frequency is at least 3.5 times the first mode frequency, so where the fundamental period exceeds $3.5 T_s$, the higher modes have larger contributions to the total response because they occur near the peak of the design response spectrum.

It follows that dynamic analysis (modal response spectrum analysis or response history analysis) may be necessary to properly evaluate buildinglike nonbuilding structures if the first mode period is larger than $3.5 T_s$ and the equivalent lateral force analysis is sufficient for nonbuilding structures that respond as single-degree-of-freedom systems.

The recommendations for nonbuilding structures provided in the following are intended to supplement the designer's judgment and experience. The designer is given considerable latitude in selecting a suitable analysis method for nonbuilding structures.

Buildinglike Nonbuilding Structures. Table 12.6-1 is used in selecting analysis methods for buildinglike nonbuilding structures, but, as illustrated in the following three conditions, the relevance of key behavior must be considered carefully:

1. Irregularities: Table 12.6-1 requires dynamic analysis for SDC D, E, and F structures that have certain horizontal or vertical irregularities. Some of these building irregularities (defined in Section 12.3.2) are relevant to nonbuilding structures. The weak- and soft-story vertical irregularities (Types 1a, 1b, 5a, and 5b of Table 12.3-2) are pertinent to the behavior of buildinglike nonbuilding structures. Other vertical and horizontal irregularities may or may not be relevant, as described below.
 - a. Horizontal irregularities: Horizontal irregularities of Types 1a and 1b affect the choice of analysis method, but these irregularities apply only where diaphragms are rigid or semirigid, and some buildinglike nonbuilding structures have either no diaphragms or flexible diaphragms.
 - b. Vertical irregularities: Vertical irregularity Type 2 is relevant where the various levels actually support significant loads. Where a buildinglike nonbuilding structure supports significant mass at a single level while other levels support small masses associated with stair landings, access platforms, and so forth, dynamic response is dominated by the first mode, so the equivalent lateral force procedure may be applied. Vertical irregularity Type 3 addresses large differences in the horizontal dimension of the seismic force-resisting system in adjacent stories because the resulting stiffness distribution can produce a fundamental mode shape unlike that assumed in the development of the equivalent lateral force procedure. Because the concern relates to stiffness distribution, the horizontal dimension of the seismic force-resisting system, not of the overall structure, is important.
2. Arrangement of supported masses: Even where a nonbuilding structure has buildinglike appearance, it may not behave like a building, depending on how masses are attached. For example, the response of nonbuilding structures with suspended vessels and boilers cannot be determined reliably using the equivalent lateral force procedure because of the pendulum modes associated with the significant mass of the suspended components. The resulting pendulum modes, while potentially reducing story shears and base shear, may require large clearances to allow pendulum motion of the supported components and may produce excessive demands on attached piping. Dynamic analysis is highly recommended in such cases, with consideration for appropriate impact forces in the absence of adequate clearances.
3. Relative rigidity of beams: Even where a classic building model may seem appropriate, the equivalent lateral force

procedure may underpredict the total response if the beams are flexible relative to the columns (of moment frames) or the braces (of braced frames). This underprediction occurs because higher modes associated with beam flexure may contribute more significantly to the total response (even if the first mode response is at a period less than $3.5 T_s$). This situation of flexible beams can be especially pronounced for nonbuilding structures because the “normal” floors common to buildings may be absent. Therefore, the dynamic analysis procedures are suggested for buildinglike nonbuilding structures with flexible beams.

Nonbuilding Structures Not Similar to Buildings. The (static) equivalent lateral force procedure is based on classic building dynamic behavior, which differs from the behavior of many nonbuilding structures not similar to buildings. As discussed below, several issues should be considered for selecting either an appropriate method of dynamic analysis or a suitable distribution of lateral forces for static analysis.

1. **Structural geometry:** The dynamic response of nonbuilding structures with a fixed base and a relatively uniform distribution of mass and stiffness, such as bottom-supported vertical vessels, stacks, and chimneys, can be represented adequately by a cantilever (shear building) model. For these structures, the equivalent lateral force procedure provided in the standard is suitable. This procedure treats the dynamic response as being dominated by the first mode. In such cases, it is necessary to identify the first mode shape (using, for instance, the Rayleigh–Ritz method or other classical methods from the literature) for distribution of the dynamic forces. For some structures, such as tanks with low height-to-diameter ratios storing granular solids, it is conservative to assume a uniform distribution of forces. Dynamic analysis is recommended for structures that have neither a uniform distribution of mass and stiffness nor an easily determined first mode shape.
2. **Number of lateral supports:** Cantilever models are obviously unsuitable for structures with multiple supports. Fig. C15.1-4 shows a nonbuilding braced frame structure that provides nonuniform horizontal support to a piece of equipment. In such cases, the analysis should include coupled model effects. For such structures, an application of the equivalent lateral force method could be used,

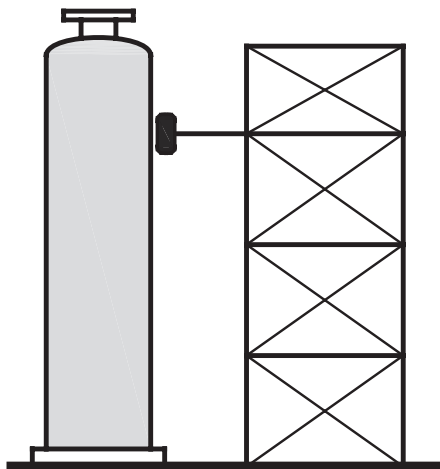


FIGURE C15.1-4 Multiple lateral supports

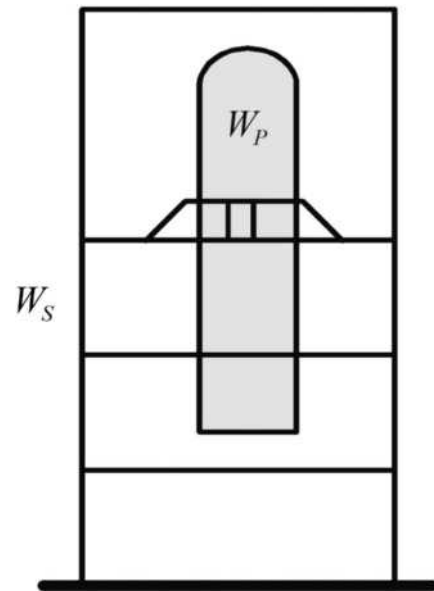


FIGURE C15.1-5 Unusual support of dead weight

- depending on the number and locations of the supports. For example, most beam-type configurations lend themselves to application of the equivalent lateral force method.
3. **Method of supporting dead weight:** Certain nonbuilding structures (such as power boilers) are supported from the top. They may be idealized as pendulums with uniform mass distribution. In contrast, a suspended platform may be idealized as a classic pendulum with concentrated mass. In either case, these types of nonbuilding structures can be analyzed adequately using the equivalent lateral force method by calculating the appropriate frequency and mode shape. Fig. C15.1-5 shows a nonbuilding structure containing lug-supported equipment with W_P greater than $0.25(W_S + W_P)$. In such cases, the analysis should include a coupled system with the mass of the equipment and the local flexibility of the supports considered in the model. Where the support is located near the nonbuilding structure’s vertical location of the center of mass, a dynamic analysis is recommended.
 4. **Mass irregularities:** Just as in the case of buildinglike nonbuilding structures, the presence of significantly uneven mass distribution is a situation where the equivalent lateral force method is not likely to provide a very accurate and perhaps unconservative force distribution. The dynamic analysis methods are recommended in such situations. Fig. C15.1-6 illustrates two such situations. In part (a), a mass irregularity exists if W_1 is greater than $1.5W_2$ or less than $0.67W_2$. In part (b), a mass irregularity exists if W_3 is greater than either $1.5W_2$ or $1.5W_4$.
 5. **Torsional irregularities:** Structures in which the fundamental mode of response is torsional or in which modes with significant mass participation exhibit a prominent torsional component may also have inertial force distributions that are significantly different from those predicted by the equivalent lateral force method. In such cases, dynamic analyses should be considered. Fig. C15.1-7 illustrates one such case where a vertical vessel is attached to a secondary vessel with W_2 greater than about $0.25(W_1 + W_2)$.
 6. **Stiffness and strength irregularities:** Just as for buildinglike nonbuilding structures, abrupt changes in the distribution

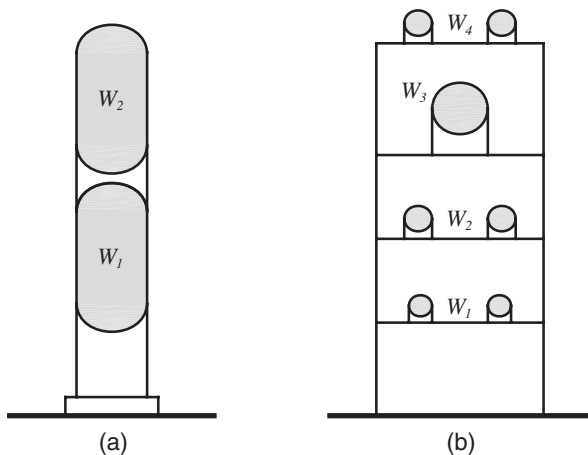


FIGURE C15.1-6 Mass irregularities

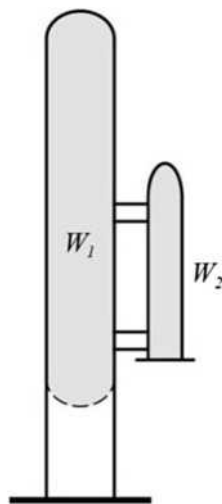


FIGURE C15.1-7 Torsional irregularity

of stiffness or strength in a nonbuilding structure not similar to buildings can result in substantially different inertial forces from those indicated by the equivalent lateral force method. Fig. C15.1-8 represents one such case. For structures that have such configurations, consideration should be given to the use of dynamic analysis procedures. Even where dynamic analysis is required, the standard does not define in any detail the degree of modeling; an adequate model may have a few dynamic degrees of freedom or tens of thousands of dynamic degrees of freedom. The important point is that the model captures the significant dynamic response features so that the resulting lateral force distribution is valid for design. The designer is responsible for determining whether dynamic analysis is warranted and, if so, the degree of detail required to address adequately the seismic performance.

7. Coupled response: Where the weight of the supported structure is large compared with the weight of the supporting structure, the combined response can be affected significantly by the flexibility of the supported nonbuilding

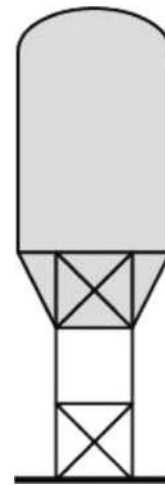


FIGURE C15.1-8 Soft-story irregularity

structure. In that case, dynamic analysis of the coupled system is recommended. Examples of such structures are shown in Fig. C15.1-9. Part (a) shows a flexible nonbuilding structure with W_p greater than $0.25(W_s + W_p)$, supported by a relatively flexible structure; the flexibility of the supports and attachments should be considered. Part (b) shows flexible equipment connected by a large-diameter, thick-walled pipe and supported by a flexible structure; the structures should be modeled as a coupled system including the pipe.

Distributed mass cantilever structures have over several cycles of ASCE 7 had their R values reduced and/or special detailing requirements added to improve their performance. The exceptions to the modal scaling rules of Section 12.9 listed in Section 15.1.3 for distributed mass cantilever structures recognize this improvement in performance.

C15.1.4 Nonbuilding Structures Sensitive to Vertical Ground Motions. Traditionally, ASCE 7 did not provide guidance to address designing for a separate vertical ground motion. Historically, this omission has not been a problem for buildings because there is inherent strength in the vertical direction because of the margin that is developed when the dead load and live load are applied. However, this is not necessarily the case for nonbuilding structures. Many nonbuilding structures are sensitive to vertical motions and do not have the benefit of the inherent strength that exists in buildings. Examples of some structures are liquid and granular storage tanks or vessels, suspended structures (such as boilers), and nonbuilding structures incorporating horizontal cantilevers. Such structures are required to incorporate Section 11.9 into the design of the structure in lieu of applying the traditional vertical ground motion of $0.2S_{DS}$.

C15.2 THIS SECTION INTENTIONALLY LEFT BLANK; SEE SECTION C15.8

C15.3 NONBUILDING STRUCTURES SUPPORTED BY OTHER STRUCTURES

There are instances where nonbuilding structures not similar to buildings are supported by other structures or other nonbuilding structures. This section specifies how the seismic design loads for

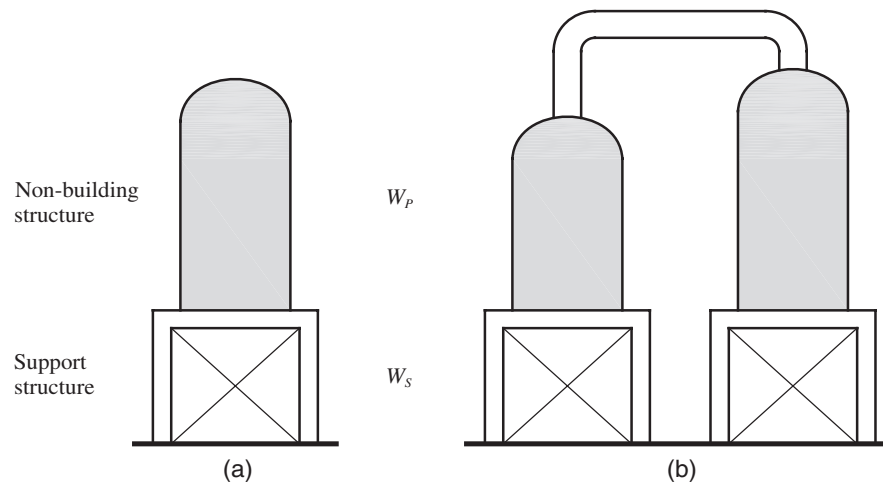


FIGURE C15.1-9 Couple system

such structures are to be determined and the detailing requirements that are to be satisfied in the design.

C15.3.1 Less Than 25% Combined Weight Condition. In many instances, the weight of the supported nonbuilding structure is relatively small compared with the weight of the supporting structure, such that the supported nonbuilding structure has a relatively small effect on the overall nonlinear earthquake response of the primary structure during design-level ground motions. It is permitted to treat such structures as nonstructural components and to use the requirements of Chapter 13 for their design. The ratio of secondary component weight to total weight of 25% at which this treatment is permitted is based on judgment and was introduced into code provisions in the 1988 *Uniform Building Code* by the SEAOC Seismology Committee. Analytical studies, typically based on linear elastic primary and secondary structures, indicate that the ratio should be lower, but the SEAOC Seismology Committee judged that the 25% ratio is appropriate where primary and secondary structures exhibit nonlinear behavior that tends to lessen the effects of resonance and interaction. In cases where a nonbuilding structure (or nonstructural component) is supported by another structure, it may be appropriate to analyze in a single model. In such cases, it is intended that seismic design loads and detailing requirements be determined following the procedures of Section 15.3.2. Where there are multiple large nonbuilding structures, such as vessels supported on a primary nonbuilding structure, and the weight of an individual supported nonbuilding structure does not exceed the 25% limit but the combined weight of the supported nonbuilding structures does, it is recommended that the combined analysis and design approach of Section 15.3.2 be used. It is also suggested that dynamic analysis be performed in such cases because the equivalent lateral force procedure may not capture some important response effects in some members of the supporting structure.

Where the weight of the supported nonbuilding structure does not exceed the 25% limit and a combined analysis is performed, the following procedure should be used to determine the F_p force of the supported nonbuilding structure based on Eq. (13.3-4):

1. A modal analysis should be performed in accordance with Section 12.9. The base shear of the combined structure and

nonbuilding structure should be taken as no less than 85% of the equivalent lateral force procedure base shear.

2. For a component supported at level i , the acceleration at that level should be taken as a_i , the total shear just below level i divided by the seismic weight at and above level i .
3. The elastic value of the component shear force coefficient should next be determined as the shear force from the modal analysis at the point of attachment of the component to the structure divided by the weight of the component. This value is preliminarily taken as $a_i a_p$. Because a_p cannot be taken as less than 1.0, the value of a_p is taken as $a_i a_p / a_i$, except that the final value a_p need not be taken as greater than 2.5 and should not be taken as less than 1.0. The final value of $a_i a_p$ should be the final value of a_i determined in step 2 multiplied by the final value of a_p determined earlier in this step.
4. The resulting value of $a_i a_p$ should be used in Eq. (13.3-4); the resulting value of F_p is subject to the maximum and minimum values of Eqs. (13.3-2) and (13.3-3), respectively.

C15.3.2 Greater Than or Equal to 25% Combined Weight Condition. Where the weight of the supported structure is relatively large compared with the weight of the supporting structure, the overall response can be affected significantly. The standard sets forth two analysis approaches, depending on the rigidity of the nonbuilding structure. The determination of what is deemed rigid or flexible is based on the same criteria used for nonstructural components.

Where the supported nonbuilding structure is rigid, it is acceptable to treat the supporting structure as a nonbuilding structure similar to a building and to determine its design loads and detailing using the requirements of Section 15.5. The design of the rigid nonbuilding structure and its anchorage is determined using the requirements of Chapter 13 with the amplification factor, a_p , taken as 1.0. However, this condition is relatively rare because the flexibility of any directly supporting members in the primary structure, such as floor beams, must be considered in determining the period of the component.

In the usual case, where the supported nonbuilding structure is flexible, a combined model of the supporting structure and the supported nonbuilding structure is used. The design loads and

detailing are determined based on the lower R value of the supported nonbuilding structure or supporting structure.

Although not specifically mentioned in Section 15.3.2, another approach is permitted. A nonlinear response history analysis of the combined system can be performed in accordance with Section 16.2, and the results can be used for the design of both the supported and supporting nonbuilding structures. This option should be considered where standard static and dynamic elastic analysis approaches may be inadequate to evaluate the earthquake response (such as for suspended boilers). This option should be used with extreme caution because modeling and interpretation of results require considerable judgment. Because of this sensitivity, Section 16.2 requires independent design review.

C15.4 STRUCTURAL DESIGN REQUIREMENTS

This section specifies the basic coefficients and minimum design forces to be used to determine seismic design loads for nonbuilding structures. It also specifies height limits and restrictions. As with building structures, it presumes that the first step in establishing the design forces is to determine the design base shear for the structure.

There are two types of nonbuilding structures: those with structural systems similar to buildings and those with structural systems not similar to buildings. Specific requirements for these two cases appear in Sections 15.5 and 15.6.

Table 15.4-1 contains the response modification coefficient (R) for nonbuilding structures similar to buildings. Table 15.4-2 contains the response modification coefficient for nonbuilding structures not similar to buildings. Every response modification coefficient has associated design and detailing requirements to ensure the required ductility associated with that response modification coefficient value (e.g., AISC 341). Some structures, such as pipe racks, do not resemble a traditional building in that they do not house people or have such things as walls and bathrooms. These structures have lateral force-resisting systems composed of braced frames and moment frames similar to a traditional building. Therefore, pipe racks are considered nonbuilding structures similar to buildings. The response modification coefficient for a pipe rack should be taken from Table 15.4-1 for the appropriate lateral force-resisting system used, and the braced frames and/or moment frames used must meet all of the design and detailing requirements associated with the R value selected (see Section 15.5.2, Pipe Racks).

Most major power distribution facility (power island) structures, such as HRSG support structures, steam turbine pedestals, coal boiler support structures, pipe racks, air inlet structures, and duct support structures, also resist lateral forces predominantly by use of buildinglike framing systems such as moment frames, braced frames, or cantilever column systems. Therefore, their response modification coefficient should be selected from Table 15.4-1, and they must meet all the design and detailing requirements associated with the response modification coefficient selected.

Many nonbuilding structures, such as flat-bottom tanks, silos, and stacks, do not use braced frames or moment frames similar to those found in buildings to resist seismic loads. Therefore, they have their own unique response modification coefficient, which can be found in Table 15.4-2.

For nonbuilding structures with lateral systems composed predominantly of buildinglike framing systems, such as moment frames, braced frames, or cantilever column systems, it would be inappropriate to extrapolate the descriptions in Table 15.4-2, resulting in inappropriately high response modification coefficients and the elimination of detailing requirements.

Once a response modification coefficient is selected from the tables, Section 15.4.1 provides additional guidance.

C15.4.1 Design Basis. Separate tables provided in this section identify the basic coefficients, associated detailing requirements, and height limits and restrictions for the two types of nonbuilding structures.

For nonbuilding structures similar to buildings, the design seismic loads are determined using the same procedures used for buildings as specified in Chapter 12, with two exceptions: fundamental periods are determined in accordance with Section 15.4.4, and Table 15.4-1 provides additional options for structural systems. Although only Section 12.8 (the equivalent lateral force procedure) is specifically mentioned in Section 15.4.1, Section 15.1.3 provides the analysis procedures that are permitted for nonbuilding structures.

In Table 15.4-1, seismic coefficients, system restrictions, and height limits are specified for a few nonbuilding structures similar to buildings. The values of R , Ω_0 , and C_d ; the detailing requirement references; and the structural system height limits are the same as those in Table 12.2-1 for the same systems, except for ordinary moment frames. In Chapter 12, increased height limits for ordinary moment frame structural systems apply to metal building systems, whereas in Chapter 15 they apply to pipe racks with end plate bolted moment connections. The seismic performance of pipe racks was judged to be similar to that of metal building structures with end plate bolted moment connections, so the height limits were made the same as those specified in previous editions.

Table 15.4-1 also provides lower R values with less restrictive height limits in SDCs D, E, and F based on good performance in past earthquakes. For some options, no seismic detailing is required if very low values of R (and corresponding high seismic design forces) are used. The concept of extending this approach to other structural systems is the subject of future research using the methodology developed in FEMA P-695 (FEMA 2009).

For nonbuilding structures not similar to buildings, the seismic design loads are determined as in Chapter 12 with three exceptions: the fundamental periods are determined in accordance with Section 15.4.4, the minima are those specified in Section 15.4.1(2), and the seismic coefficients are those specified in Table 15.4-2.

Some entries in Table 15.4-2 may seem to be conflicting or confusing. For example, the first major entry is for elevated tanks, vessels, bins, or hoppers. A subset of this entry is for tanks on braced or unbraced legs. This subentry is intended for structures where the supporting columns are integral with the shell (such as an elevated water tank). Tension-only bracing is allowed for such a structure. Where the tank or vessel is supported by buildinglike frames, the frames are to be designed in accordance with all of the restrictions normally applied to building frames. Section 15.3 provides provisions for nonbuilding structures supported by buildinglike frames. Beginning with the 2005 edition of ASCE 7, Table 15.4-2 contained an entry for "Tanks or vessels supported on structural towers similar to buildings." Under certain circumstances, text provided with this table entry conflicted with the requirements of Section 15.3. If the weight of the nonbuilding structure is relatively small compared to the weight of the structure (less than 25% of the weight of the structure) or the nonbuilding structure is rigid, the supported nonbuilding structure can be treated as a nonstructural component and the values of the supporting structure seismic coefficients can be taken from Table 15.4-1. Under these circumstances, the deleted entry was correct. However, if the weight of the supported nonbuilding structure is not small and the

nonbuilding structure is flexible (which is generally the case especially when you consider the vertical and rocking flexibility of supporting floor beams), the seismic coefficients are determined as the most conservative.

The accidental torsion requirements of Section 12.8.4.2 were formulated primarily for use in building structures. The primary factors that contribute to accidental torsion are lateral force-resisting systems that are located primarily near the center of the structure rather than the perimeter, disproportionate concentration of inelastic demands in system components, the effects of nonstructural elements, uncertainties in defining the structure's stiffness characteristics, and spatial variation (and rotational components of ground motions) of horizontal input motions applied to long structures. Inherently torsionally resistant systems as defined in Section 15.4.1, Item 5, with R values less than or equal to 3.5 are not expected to have inelastic demands of a level that would require additional consideration of accidental torsion. Additionally, nonbuilding structures rarely contain significant nonstructural elements that are not accounted for explicitly in the design of these structures and typically have very well-known mass and stiffness characteristics. Nonbuilding structures also rarely, if ever, have their lateral force-resisting systems located at the center of the structure in plan rather than at the perimeter. The requirement that the calculated center of rigidity at each diaphragm is greater than 5% of the plan dimension of the diaphragm in each direction from the calculated center of mass of the diaphragm prevents configurations of lateral force-resisting elements that are inherently susceptible to the effects of torsion from being exempted from the effects of accidental torsion. Spatial variations of ground motions should be considered in the design of structures of considerable length. If there are significant variations between full and empty weights of the structure, the inherent torsion caused by these variations should be considered in the design of the structure. If there is a nonuniform distribution of mass in silos or bins storing bulk materials because of multiple filling or discharge points, multiple hoppers, nonuniform funnel flow, bulk material behavior, or other operational considerations, the inherent torsion caused by these conditions should be considered in the design of the silo or bin.

C15.4.1.1 Importance Factor. The Importance Factor for a nonbuilding structure is based on the risk category defined in Chapter 1 of the standard or the building code being used in conjunction with the standard. In some cases, reference standards provide a higher Importance Factor, in which case the higher Importance Factor is used.

If the Importance Factor is taken as 1.0 based on a hazard and operability (HAZOP) analysis performed in accordance with Chapter 1, the third paragraph of Section 1.5.3 requires careful consideration; worst-case scenarios (instantaneous release of a vessel or piping system) must be considered. HAZOP risk analysis consultants often do not make such assumptions, so the design professional should review the HAZOP analysis with the HAZOP consultant to confirm that such assumptions have been made to validate adjustment of the Importance Factor. Clients may not be aware that HAZOP consultants do not normally consider the worst-case scenario of instantaneous release but tend to focus on other, more hypothetical, limited-release scenarios, such as those associated with a 2-in.² (1,290 mm²) hole in a tank or vessel.

C15.4.2 Rigid Nonbuilding Structures. The definition of rigid (having a natural period of less than 0.06 s) was selected judgmentally. Below that period, the energy content of seismic ground motion is generally believed to be very low, and therefore the building response is not likely to be excessively amplified.

Also, it is unlikely that any building will have a first mode period as low as 0.06 s, and it is even unusual for a second mode period to be that low. Thus, the likelihood of either resonant behavior or excessive amplification becomes quite small for equipment that has periods below 0.06 s.

The analysis to determine the period of the nonbuilding structure should include the flexibility of the soil subgrade.

C15.4.3 Loads. As for buildings, the seismic weight must include the range of design operating weight of permanent equipment.

C15.4.4 Fundamental Period. A significant difference between building structures and nonbuilding structures is that the approximate period formulas and limits of Section 12.8.2.1 may not be used for nonbuilding structures. In lieu of calculating a specific period for a nonbuilding structure for determining seismic lateral forces, it is of course conservative to assume a period of $T = T_s$, which results in the largest lateral design forces. Computing the fundamental period is not considered a significant burden because most commonly used computer analysis programs can perform the required calculations.

C15.4.7 Drift, Deflection, and Structure Separation. Nonbuilding structure drifts, deflection, and structure separation are calculated using strength design factored load combinations in order to be compatible with the seismic load definition and the definition of the C_d factors. This philosophy is consistent with that of drift, deflections, and structure separation for buildings defined in Chapter 12.

C15.4.8 Site-Specific Response Spectra. Where site-specific response spectra are required, they should be developed in accordance with Chapter 21 of the standard. If determined for other recurrence intervals, Section 21.1 applies, but Sections 21.2 through 21.4 apply only to risk-targeted maximum considered earthquake (MCE_R) determinations. Where other recurrence intervals are used, it should be demonstrated that the requirements of Chapter 15 also are satisfied.

C15.4.9 Anchors in Concrete or Masonry. Many nonbuilding structures rely on the ductile behavior of anchor bolts to justify the response modification factor, R , assigned to the structure. Nonbuilding structures typically rely more heavily on anchorage to provide system ductility. The additional requirements of Section 15.4.9 provide additional anchorage strength and ductility to support the response modification factors assigned to these systems. The addition of Section 15.4.9 provides a consistent treatment of anchorage for nonbuilding structures.

C15.4.9.4 ASTM F1554 Anchors. ASTM F1554 contains a requirement that is not consistent with the anchor requirements found in Chapter 15. Section 6.4 of ASTM F1554 allows the anchor supplier to substitute weldable Grade 55 anchors for Grade 36 anchors without the approval of the registered design professional. Because many nonbuilding structures rely on the ability of the anchors to stretch to justify the response modification factor, R , assigned to the structure, a higher yield anchor cannot be allowed to be substituted for a lower yield anchor without the approval of the registered design professional. Except where anchors are specified and are designed as ductile steel anchors in accordance with ACI 318, Section 17.2.3.4.3(a), or where the design must meet the requirements of Section 15.7.5 or Section 15.7.11.7b, this provision does not prohibit ductility from being provided by another element of the structure. In that case, the ASTM F1554 anchors would be designed for the corresponding forces.

C15.4.10 Requirements for Nonbuilding Structure Foundations on Liquefiable Sites. Section 12.13.9 allows shallow foundation to be built on liquefiable soils with a number of restrictions. Many nonbuilding structures are sensitive to large foundation settlements. This sensitivity is caused by restraint imposed by interconnecting piping and equipment and the buckling sensitivity of shell structures. Therefore, in order to build these structures on shallow foundations on liquefiable soils, it must be demonstrated that the foundation, nonbuilding structure not similar to buildings, and connecting systems can be designed for the soil strength loss, the anticipated settlements from lateral spreading, and total and differential settlements induced by MCE_G earthquake ground motions.

C15.5 NONBUILDING STRUCTURES SIMILAR TO BUILDINGS

C15.5.1 General. Although certain nonbuilding structures exhibit behavior similar to that of building structures, their functions and occupancies are different. Section 15.5 of the standard addresses the differences.

C15.5.2 Pipe Racks. Freestanding pipe racks supported at or below grade with framing systems that are similar to building systems are designed in accordance with Section 12.8 or 12.9 and Section 15.4. Single-column pipe racks that resist lateral loads should be designed as inverted pendulums.

Based on good performance in past earthquakes, Table 15.4-1 sets forth the option of lower R values and less restrictive height limits for structural systems commonly used in pipe racks. The R value versus height limit tradeoff recognizes that the size of some nonbuilding structures is determined by factors other than traditional loadings and results in structures that are much stronger than required for seismic loadings. Therefore, the ductility demand is generally much lower than that for a corresponding building. The intent is to obtain the same structural performance at the increased heights. This option proves to be economical in most situations because of the relative cost of materials and construction labor. The lower R values and increased height limits of Table 15.4-1 apply to nonbuilding structures similar to buildings; they cannot be applied to building structures. Table C15.5-1 illustrates the R values and height limits for a 70-ft (21.3-m) high steel ordinary moment frame (OMF) pipe rack.

C15.5.3.1 Steel Storage Racks. The two approaches to the design of steel storage racks set forth by the standard are intended to produce comparable results. The specific revisions to the Rack Manufacturers Institute (RMI) specification cited in earlier

editions of this standard and the detailed requirements of the ANSI/RMI MH 16.1 specification reflect the recommendations of FEMA 460 (FEMA 2005).

Although the ANSI/RMI MH 16.1 specification reflects the recommendations of FEMA 460 (FEMA 2005), the anchorage provisions of the ANSI/RMI MH 16.1 specification are not in conformance with ASCE/SEI 7. Therefore, specific anchorage requirements were added in Sections 15.5.3.1.1 and 15.5.3.1.2.

These recommendations address the concern that storage racks in warehouse-type retail stores may pose a greater seismic risk to the general public than exists in low-occupancy warehouses or more conventional retail environments. Under normal conditions, retail stores have a far higher occupant load than an ordinary warehouse of a comparable size. Failure of a storage rack system in a retail environment is much more likely to cause personal injury than would a similar failure in a storage warehouse. To provide an appropriate level of additional safety in areas open to the public, an Importance Factor of 1.50 is specified. Storage rack contents, though beyond the scope of the standard, may pose a potentially serious threat to life should they fall from the shelves in an earthquake. It is recommended that restraints be provided, as shown in Fig. C15.5-1, to prevent the contents of rack shelving open to the general public from falling during strong ground shaking.

C15.5.3.2 Steel Cantilevered Storage Racks. The two approaches to the design of steel cantilevered storage racks set forth by the standard are intended to produce comparable results. The specific development of a new RMI standard to include the detailed requirements of the new ANSI/RMI MH 16.3 (2016) specification, reflect the unique characteristics of this structural storage system, along with the recommendations of FEMA 460, *Seismic Considerations for Steel Storage Racks Located in Areas Accessible to the Public*.

The values of R , C_d , and Ω_0 added to Table 15.4-1 for Steel Cantilever Storage Racks were taken directly from Table 2.7.2.2.3 (1) of ANSI/RMI MH 16.3.

The anchorage provisions of the ANSI/RMI MH 16.3 specification are not in conformance with ASCE/SEI 7. Therefore, specific anchorage requirements were added in Section 15.5.3.2.1.

These recommendations address the concern that steel cantilevered storage racks in warehouse-type retail stores may pose a greater seismic risk to the general public than exists in low-occupancy warehouses or more conventional retail

Table C15.5-1 R Value Selection Example for Steel OMF Pipe Racks

Seismic Design Category	R	ASCE/SEI 7-10 Table	System	Seismic Detailing Requirements
C	3.5	12.2-1 or 15.4-1	Steel ordinary moment frame (OMF)	AISC 341
C	3	12.2-1	Structural steel systems not specifically detailed for seismic resistance	None
D or E	2.5	15.4-1	Steel OMF with permitted height increase	AISC 341
D, E, or F	1	15.4-1	Steel OMF with unlimited height	AISC 341



FIGURE C15.5-1 Merchandise restrained by netting

Source: FEMA 460 *Seismic Considerations for Steel Storage Racks*.



FIGURE C15.5-2 Examples of Structural Towers That are Not Integral with the Supported Tank

Source: (left) Courtesy of Chevron; reproduced with permission. (right) Courtesy of CB&I LLC; reproduced with permission.

environments. Under normal conditions, retail stores have a far higher occupant load than an ordinary warehouse of a comparable size. Failure of a steel cantilevered storage rack system in a retail environment is much more likely to cause personal injury than would a similar failure in a storage warehouse. To provide an appropriate level of additional safety in areas open to the public, an Importance Factor of 1.50 is specified. Steel cantilevered storage rack contents, though beyond the scope of the standard, may pose a potentially serious threat to life should they fall from the shelves in an earthquake. It is recommended that restraints be provided, as shown in Figure C15.5-1, to prevent the contents of rack shelving open to the general public from falling during strong ground shaking.

All systems in ANSI/MH16.3, Table 2.7.2.2.3(1) are ordinary systems. For all systems in SDC B and C, the values in ANSI/MH16.3 (2016), Table 2.7.2.2.3(1) for R , Ω_0 , and C_d correspond to the values shown in Table 12.2-1 for Steel Systems Not Specifically Detailed for Seismic Resistance, Excluding Cantilever Column Systems. No seismic detailing is required. For hot-rolled steel systems in SDC D, E, and F, the values in ANSI/MH16.3, Table 2.7.2.2.3(1) for R , Ω_0 , and C_d correspond to the values shown in Table 15.4-1 for ordinary systems with permitted height increase except that no height limits apply. The hot-rolled steel systems are detailed to AISC 341. For cold-formed steel systems in SDC D, E, and F, the values in ANSI/MH16.3 (2016), Table 2.7.2.2.3(1) for R , Ω_0 , and C_d correspond to the values shown in Table 15.4-1 for ordinary systems with unlimited height. Seismic detailing is not required for the cold-formed steel systems.

C15.5.4 Electrical Power-Generating Facilities. Electrical power plants closely resemble building structures, and their performance in seismic events has been good. For reasons of mechanical performance, lateral drift of the structure must be limited. The lateral bracing system of choice has been the concentrically braced frame. In the past, the height limits on braced frames in particular have been an encumbrance to the design of large power-generating facilities. Based on acceptable past performance, Table 15.4-1 permits the use of ordinary concentrically braced frames with both lower R values and less restrictive height limits. This option is particularly effective for boiler buildings, which generally are 300 ft (91.4 m) or more high. A peculiarity of large boiler buildings is the general practice of suspending the boiler from the roof structures; this practice results in an unusual mass distribution, as discussed in Section C15.1.3.



FIGURE C15.5-3 Examples of Structural Towers That are Integral with the Supported Tank

Source: Courtesy of CB&I LLC; reproduced with permission.

C15.5.5 Structural Towers for Tanks and Vessels. The requirements of this section apply to structural towers that are not integral with the supported tank. Elevated water tanks designed in accordance with AWWA D100 are not subject to Section 15.5.5. A structural tower supporting a tank or vessel is considered integral with the supported tank or vessel where the tank or vessel shell acts as a part of the seismic force-resisting system of the supporting tower.

Examples of structural towers that are not integral with the supported tank are shown in Fig. C15.5-2. Examples of structural towers that are integral with the supported tank are shown in Fig. C15.5-3. Examples of structural towers that are integral with the supported tank include column-supported elevated water tanks designed to AWWA D100 and column-supported liquid and gas spheres designed to ASME BVPC, Section VIII.

C15.5.6 Piers and Wharves. Current industry practice recognizes the distinct differences between the two categories of piers and wharves described in the standard. Piers and wharves with public occupancy, described in Section 15.5.6.2, are commonly treated as the “foundation” for buildings or buildinglike structures; design is performed using the standard, likely under the jurisdiction of the local building official. Piers and wharves without occupancy by the general public are often treated differently and are outside the scope of the standard; in many cases, these structures do not fall under the jurisdiction of

building officials, and design is performed using other industry-accepted approaches.

Design decisions associated with these structures often reflect economic considerations by both owners and local, regional, or state jurisdictional entities with interest in commercial development. Where building officials have jurisdiction but lack experience analyzing pier and wharf structures, reliance on other industry-accepted design approaches is common.

Where occupancy by the general public is not a consideration, seismic design of structures at major ports and marine terminals often uses a performance-based approach, with criteria and methods that are very different from those used for buildings, as provided in the standard. Design approaches most commonly used are generally consistent with the practices and criteria described in the following documents: *Seismic Design Guidelines for Port Structures* (2001); Ferritto et al. (1999); Priestley et al. (1996); Werner (1998); *Marine Oil Terminal Engineering and Maintenance Standards* (2005).

These alternative approaches have been developed over a period of many years by working groups within the industry, and they reflect the historical experience and performance characteristics of these structures, which are very different from those of building structures.

The main emphasis of the performance-based design approach is to provide criteria and methods that depend on the economic importance of a facility. Adherence to the performance criteria in the documents listed does not seek to provide uniform margins of collapse for all structures; their application is expected to provide at least as much inherent life safety as for buildings designed using the standard. The reasons for the higher inherent level of life safety for these structures include the following:

1. These structures have relatively infrequent occupancy, with few working personnel and very low density of personnel. Most of these structures consist primarily of open area, with no enclosed structures that can collapse onto personnel. Small control buildings on marine oil terminals or similar secondary structures are commonly designed in accordance with the local building code.
2. These pier or wharf structures typically are constructed of reinforced concrete, prestressed concrete, or steel and are highly redundant because of the large number of piles supporting a single wharf deck unit. Tests done at the University of California at San Diego for the Port of Los Angeles have shown that high ductilities (10 or more) can be achieved in the design of these structures using practices currently used in California ports.
3. Container cranes, loading arms, and other major structures or equipment on piers or wharves are specifically designed not to collapse in an earthquake. Typically, additional piles and structural members are incorporated into the wharf or pier specifically to support such items.
4. Experience has shown that seismic “failure” of wharf structures in zones of strong seismicity is indicated not by collapse but by economically irreparable deformations of the piles. The wharf deck generally remains level or slightly tilting but shifts out of position. Earthquake loading on properly maintained marine structures has never induced complete failure that could endanger life safety.
5. The performance-based criteria of the listed documents address reparability of the structure. These criteria are much more stringent than collapse prevention criteria and create a greater margin for life safety.

Lateral load design of these structures in low, or even moderate, seismic regions often is governed by other marine conditions.

C15.6 GENERAL REQUIREMENTS FOR NONBUILDING STRUCTURES NOT SIMILAR TO BUILDINGS

Nonbuilding structures not similar to buildings exhibit behavior markedly different from that of building structures. Most of these types of structures have reference documents that address their unique structural performance and behavior. The ground motion in the standard requires appropriate translation to allow use with industry standards.

C15.6.1 Earth-Retaining Structures. Section C11.8.3 presents commonly used approaches for the design of nonyielding walls and yielding walls for bending, overturning, and sliding, taking into account the varying soil types, importance, and site seismicity.

C15.6.2 Chimneys and Stacks

C15.6.2.1 General. The design of stacks and chimneys to resist natural hazards generally is governed by wind design considerations. The exceptions to this general rule involve locations with high seismicity, stacks and chimneys with large elevated masses, and stacks and chimneys with unusual geometries. It is prudent to evaluate the effect of seismic loads in all but those areas with the lowest seismicity. Although not specifically required, it is recommended that the special seismic details required elsewhere in the standard be considered for application to stacks and chimneys.

C15.6.2.2 Concrete Chimneys and Stacks. Concrete chimneys typically possess low ductility, and their performance is especially critical in the regions around large (breach) openings because of reductions in strength and loss of confinement for vertical reinforcement in the jamb regions around the openings. Earthquake-induced chimney failures have occurred in recent history (in Turkey in 1999) and have been attributed to strength and detailing problems (Kilic and Sozen 2003). Therefore, the R value of 3 traditionally used in ASCE/SEI 7-05 for concrete stacks and chimneys was reduced to 2, and detailing requirements for breach openings were added in the 2010 edition of this standard.

C15.6.2.3 Steel Chimneys and Stacks. Guyed steel stacks and chimneys generally are lightweight. As a result, the design loads caused by natural hazards generally are governed by wind. On occasion, large flares or other elevated masses located near the top may require in-depth seismic analysis. Although it does not specifically address seismic loading, Chapter 6 of Troitsky (1990) provides a methodology appropriate for resolution of the seismic forces defined in the standard in addition to the requirements found in ASME STS-1.

C15.6.4 Special Hydraulic Structures. The most common special hydraulic structures are baffle walls and weirs that are used in water treatment and wastewater treatment plants. Because there are openings in the walls, during normal operations the fluid levels are equal on each side of the wall, exerting no net horizontal force. Sloshing during a seismic event can exert large forces on the wall, as illustrated in Fig. C15.6-1. The walls can fail unless they are designed properly to resist the dynamic fluid forces.

C15.6.5 Secondary Containment Systems. This section reflects the judgment that designing all impoundment dikes for the MCE_R ground motion when full and sizing all impoundment dikes for the sloshing liquid height is too conservative. Designing an impoundment dike as full for the MCE_R assumes failure of the

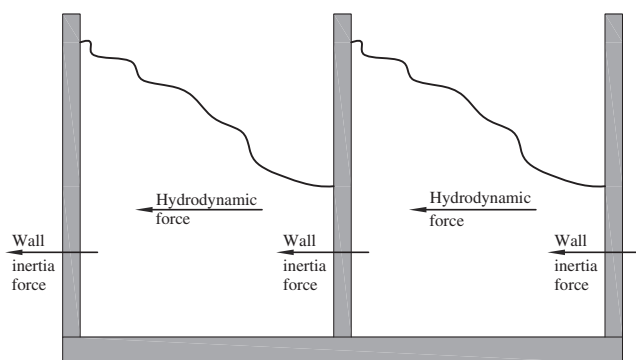


FIGURE C15.6-1 Wall forces

primary containment and occurrence of a significant aftershock. Such significant aftershocks (of the same magnitude as the MCE_R ground motion) are rare and do not occur in all locations. Although explicit design for aftershocks is not a requirement of the standard, secondary containment must be designed full for an aftershock to protect the general public. The use of two-thirds of the MCE_R ground motion as the magnitude of the design aftershock is supported by Bath's law, according to which the maximum expected aftershock magnitude may be estimated to be 1.2 scale units below the main shock magnitude.

The risk assessment and risk management plan described in Section 1.5.2 are used to determine where the secondary containment must be designed full for the MCE_R . The decision to design secondary containment for this more severe condition should be based on the likelihood of a significant aftershock occurring at the particular site, considering the risk posed to the general public by the release of hazardous material from the secondary containment.

Secondary containment systems must be designed to contain the sloshing liquid height where the release of liquid would place the general public at risk by exposing them to hazardous materials, by scouring of foundations of adjacent structures, or by causing other damage to adjacent structures.

C15.6.5.1 Freeboard. Eq. (15.6-1) was revised in ASCE 7-10 to return to the more exact theoretical formulation for sloshing liquid height instead of the rounded value introduced in ASCE/SEI 7-05. The rounded value in part accounted for maximum direction of response effects. Because the ground motion definition in ASCE/SEI 7-10 was changed and the maximum direction of response is now directly accounted for, it is no longer necessary to account for these effects by rounding up the theoretical sloshing liquid height factor in Eq. (15.6-1).

C15.6.6 Telecommunication Towers. Telecommunication towers support small masses, and their design generally is governed by wind forces. Although telecommunication towers have a history of experiencing seismic events without failure or significant damage, seismic design in accordance with the standard is required.

Typically bracing elements bolt directly (without gusset plates) to the tower legs, which consist of pipes or bent plates in a triangular plan configuration.

C15.6.7 Steel Tubular Support Structures for Onshore Wind Turbine Generator Systems. The most common support structures for large onshore wind turbine generator systems are steel tubular towers. Recommendations for the design of these structures can be found in ASCE/AWEA (2011). ASCE/AWEA (2011) applies to wind turbines that have a rotor-swept area



FIGURE C15.6-2 Typical Steel Tubular Support Structure for Onshore Wind Turbine Generator Systems

Source: Courtesy of GE Power; reproduced with permission.

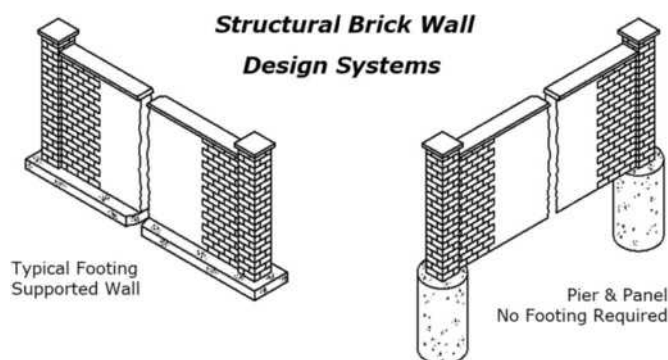


FIGURE C15.6-3 Typical Cantilever Wall Systems Falling under the Requirements of Section 15.6.8.

Source: Courtesy of J.G. Soules; reproduced with permission.

greater than 2,153 ft² (200 m²). These recommendations are to be used in conjunction with seismic lateral forces determined in accordance with Section 15.4. A typical steel tubular support structure for an onshore wind turbine generator system is shown in Fig. C15.6-2.

C15.6.8 Ground-Supported Cantilever Walls or Fences. Ground-supported cantilever walls and fences constructed from masonry, concrete, timber, or a combination of materials, including steel, are common. Such walls are often used as sound barrier walls or to limit access to residential subdivisions. Ground-supported cantilever walls and fences include walls supported by a footing and pier and panel/pilaster and panel wall systems (Fig. C15.6-3) as long as these systems are not supported laterally in the out-of-plane direction above grade. An example of a masonry ground-supported cantilever wall is shown in Fig. C15.6-4. Many improperly designed ground-supported cantilever walls and fences constructed from masonry or concrete have experienced problems and have failed during seismic events



FIGURE C15.6-4 Typical Masonry Ground-Supported Cantilever Wall

as evidenced in Section 6.3.9.1 of FEMA E-74 (2012), *Reducing the Risks of Nonstructural Earthquake Damage—A Practical Guide*.

The provisions for ground-supported cantilever walls and fences more than 6 ft (1.83 m) high were contained in prior issues of the Uniform Building Code, including the 1997 Uniform Building Code (ICBO 1997). When the International Building Code was developed, the provisions were inadvertently dropped and were not incorporated in ASCE 7. Walls of all heights should be properly designed. The 6-ft (1.83-m) height has been retained from the 1997 Uniform Building Code as the minimum height at which these provisions apply because walls less than 6 ft (1.83 m) high are not deemed to present as significant a risk to life safety.

The seismic design parameters chosen for this system are based on those given in Table 15.4-2 for “all other self-supporting structures, tanks, or vessels not covered above or by reference standards that are not similar to buildings” except that all height limits were changed to no limit (NL), considering that the structure is a cantilever wall. Cantilever walls covered by these provisions can be of any material or combination of materials; therefore, a relatively low value of R was chosen to account for these material combinations. Additionally, pilasters incorporated in many of these wall systems are essentially ordinary cantilever columns. Ordinary cantilever columns in ASCE 7 tend to have low R values irrespective of the material used.

A decision was made by the ASCE 7 Seismic Subcommittee that a ground-supported freestanding wall or fence was a nonbuilding structure not similar to a building and should fall under the provisions of Chapter 15 instead of Chapter 13.

C15.7 TANKS AND VESSELS

C15.7.1 General. Methods for seismic design of tanks, currently adopted by a number of reference documents, have evolved from earlier analytical work by Jacobsen (1949), Housner (1963), Velestos (1974), Haroun and Housner (1981), and others. The procedures used to design flat-bottom storage tanks and liquid containers are based on the work of Housner (U.S. Department of Energy, 1963) and Wozniak and Mitchell (1978). The reference documents for tanks and vessels have specific requirements to safeguard against catastrophic failure of the primary structure

based on observed behavior in seismic events since the 1930s. Other methods of analysis, using flexible shell models, have been proposed but at present are beyond the scope of the standard.

The industry-accepted design methods use three basic steps:

1. Dynamic modeling of the structure and its contents. When a liquid-filled tank is subjected to ground acceleration, the lower portion of the contained liquid, identified as the impulsive component of mass, W_i , acts as if it were a solid mass rigidly attached to the tank wall. As this mass accelerates, it exerts a horizontal force, P_i , on the wall; this force is directly proportional to the maximum acceleration of the tank base. This force is superimposed on the inertia force of the accelerating wall itself, P_s . Under the influence of the same ground acceleration, the upper portion of the contained liquid responds as if it were a solid mass flexibly attached to the tank wall. This portion, which oscillates at its own natural frequency, is identified as the convective component, W_c , and exerts a horizontal force, P_c , on the wall. The convective component oscillations are characterized by sloshing whereby the liquid surface rises above the static level on one side of the tank and drops below that level on the other side.
2. Determination of the period of vibration, T_i , of the tank structure and the impulsive component and determination of the natural period of oscillation (sloshing), T_c , of the convective component.
3. Selection of the design response spectrum. The response spectrum may be site specific, or it may be constructed on the basis of seismic coefficients given in national codes and standards. Once the design response spectrum is constructed, the spectral accelerations corresponding to T_i and T_c are obtained and are used to calculate the dynamic forces P_i , P_s , and P_c .

Detailed guidelines for the seismic design of circular tanks, incorporating these concepts to varying degrees, have been the province of at least four industry reference documents: AWWA D100 for welded steel tanks (since 1964); API 650 for petroleum storage tanks; AWWA D110 for prestressed, wire-wrapped tanks (since 1986); and AWWA D115 for prestressed concrete tanks stressed with tendons (since 1995). In addition, API 650 and API 620 contain provisions for petroleum, petrochemical, and cryogenic storage tanks. The detail and rigor of analysis prescribed in these documents have evolved from a semistatic approach in the early editions to a more rigorous approach at present, reflecting the need to include the dynamic properties of these structures.

The requirements in Section 15.7 are intended to link the latest procedures for determining design-level seismic loads with the allowable stress design procedures based on the methods in the standard. These requirements, which in many cases identify specific substitutions to be made in the design equations of the reference documents, will assist users of the standard in making consistent interpretations.

ACI has published ACI 350.3-06 (2006), *Seismic Design of Liquid-Containing Concrete Structures*. This document, which addresses all types of concrete tanks (prestressed and nonprestressed, circular, and rectilinear), has provisions that are unfortunately not consistent with the seismic criteria of ASCE/SEI 7. However, the document, when combined with the modifications required in Section 15.7.7.3, serves as both a practical “how-to” loading reference and a guide to supplement application of ACI 318, Chapter 18.

C15.7.2 Design Basis. In the case of the seismic design of nonbuilding structures, standardization requires adjustments to

industry reference documents to minimize existing inconsistencies among them, while recognizing that structures designed and built over the years in accordance with these documents have performed well in earthquakes of varying severity. Of the inconsistencies among reference documents, the ones most important to seismic design relate to the base shear equation. The traditional base shear takes the following form:

$$V = \frac{ZIS}{R_w} CW \quad (\text{C15.7-1})$$

An examination of those terms as used in the different references reveals the following:

1. Z , S : The seismic zone coefficient, Z , has been rather consistent among all the documents because it usually has been obtained from the seismic zone designations and maps in the model building codes. However, the soil profile coefficient, S , does vary from one document to another. In some documents, these two terms are combined.
2. I : The Importance Factor, I , has varied from one document to another, but this variation is unavoidable and understandable because of the multitude of uses and degrees of importance of tanks and vessels.
3. C : The coefficient C represents the dynamic amplification factor that defines the shape of the design response spectrum for any given ground acceleration. Because C is primarily a function of the frequency of vibration, inconsistencies in its derivation from one document to another stem from at least two sources: differences in the equations for the determination of the natural frequency of vibration, and differences in the equation for the coefficient itself. (For example, for the shell/impulsive liquid component of lateral force, the steel tank documents use a constant design spectral acceleration [constant C] that is independent of the "impulsive" period, T .) In addition, the value of C varies depending on the damping ratio assumed for the vibrating structure (usually between 2% and 7% of critical).
4. Where a site-specific response spectrum is available, calculation of the coefficient C is not necessary except in the case of the convective component (coefficient C_c), which is assumed to oscillate with 0.5% of critical damping and whose period of oscillation is usually long (greater than 2.5 s). Because site-specific spectra are usually constructed for high damping values (3% to 7% of critical) and because the site-specific spectral profile may not be well-defined in the long-period range, an equation for C_c applicable to a 0.5% damping ratio is necessary to calculate the convective component of the seismic force.
5. R_w : The response modification factor, R_w , is perhaps the most difficult to quantify, for a number of reasons. Although R_w is a compound coefficient that is supposed to reflect the ductility, energy-dissipating capacity, and redundancy of the structure, it is also influenced by serviceability considerations, particularly in the case of liquid-containing structures.

In the standard, the base shear equation for most structures has been reduced to $V = C_s W$, where the seismic response coefficient, C_s , replaces the product ZSC/R_w . C_s is determined from the design spectral response acceleration parameters S_{DS} and S_{D1} (at short periods and at a period of 1, respectively), which in turn are obtained from the mapped MCE_R spectral accelerations S_s and S_1 . As in the case of the prevailing industry reference

documents, where a site-specific response spectrum is available, C_s is replaced by the actual values of that spectrum.

The standard contains several bridging equations, each designed to allow proper application of the design criteria of a particular reference document in the context of the standard. These bridging equations associated with particular types of liquid-containing structures and the corresponding reference documents are discussed below. Calculation of the periods of vibration of the impulsive and convective components is in accordance with the reference documents, and the detailed resistance and allowable stresses for structural elements of each industry structure are unchanged, except where new information has led to additional requirements.

It is expected that the bridging equations of Section 15.7.7.3 will be eliminated as the relevant reference documents are updated to conform to the standard. The bridging equations previously provided for AWWA D100 and API 650 already have been eliminated as a result of updates of these documents.

Tanks and vessels are sensitive to vertical ground motions. Traditionally, the approach has been to apply a vertical seismic coefficient equal to $0.2S_{DS}$ to the design. This design approach came from the process used to design buildings and may underestimate the vertical response of the tank and its contents. For noncylindrical tanks, the increase in the hydrostatic pressure caused by vertical excitation has taken the form of $0.4S_{av}$, where S_{av} is determined in accordance with Section 15.7.2 and Section 11.9. This pressure is combined directly with the hydrodynamic loads induced from lateral ground motions. The result is equal to 100% horizontal plus 40% vertical. The response of cylindrical tanks to vertical motions is well known and documented in various papers. Unless otherwise specified in a reference document, the vertical period T_v , may be determined by

$$T_v = 2\pi \sqrt{\frac{\gamma_L R H_L^2}{g t E}} \quad (\text{C15.7-2})$$

where

- γ_L = Unit weight of stored liquid;
- R = Tank radius to the inside of the wall;
- H_L = liquid height inside the tank;
- g = acceleration caused by gravity in consistent units;
- t = average shell thickness; and
- E = modulus of elasticity of shell.

Eq. (C15.7-2) comes from ACI 350.3 (2006) and is based on a rigid response of the liquid to vertical ground motions. Additional documents, such as Section 7.7.1 of ASCE's *Guidelines for the Seismic Design of Oil and Gas Pipeline Systems* (1984) provide solutions to determine the response of a flexible tank to vertical ground motions. The response of the structure itself is set equal to 0.4 times the peak of the vertical response spectra. Using the peak of the vertical response spectra recognizes the vertical stiffness of the tank walls. This load is combined directly with loads produced from lateral ground motions. The result is equal to 100% horizontal plus 40% vertical. It should also be noted that R has been added to Eq. (15.7-1). R is included in the calculation of hoop stress because the response of the tank shell caused by the added hoop tension from vertical ground motions is no different than the response of the tank shell caused by the added hoop tension from horizontal ground motions. ACI 350.3 has used this philosophy for many years.

C15.7.3 Strength and Ductility. As is the case for building structures, ductility and redundancy in the lateral support systems

for tanks and vessels are desirable and necessary for good seismic performance. Tanks and vessels are not highly redundant structural systems, and therefore ductile materials and well-designed connection details are needed to increase the capacity of the vessel to absorb more energy without failure. The critical performance of many tanks and vessels is governed by shell stability requirements rather than by yielding of the structural elements. For example, contrary to building structures, ductile stretching of anchor bolts is a desirable energy absorption component where tanks and vessels are anchored. The performance of cross-braced towers is highly dependent on the ability of the horizontal compression struts and connection details to develop fully the tension yielding in the rods. In such cases, it is also important to preclude both premature failure in the threaded portion of the connection and failure of the connection of the rod to the column before yielding of the rod.

The changes made to Section 15.7.3(a) are intended to ensure that anchors and anchor attachments are designed such that the anchor yields (stretches) before the anchor attachment to the structure fails. The changes also clarify that the anchor rod embedment requirements are to be based on the requirements of Section 15.7.5 and not Section 15.7.3(a).

C15.7.4 Flexibility of Piping Attachments. Poor performance of piping connections (tank leakage and damage) caused by seismic deformations is a primary weakness observed in seismic events. Although commonly used piping connections can impart mechanical loads to the tank shell, proper design in seismic areas results in only negligible mechanical loads on tank connections subject to the displacements shown in Table 15.7-1. API 650 treats the values shown in Table 15.7-1 as allowable stress-based values and therefore requires that these values be multiplied by 1.4 where strength-based capacity values are required for design.

The displacements shown in Table 15.7-1 are based on movements observed during past seismic events. The vertical tank movements listed are caused by stretch of the mechanical anchors or steel tendons (in the case of a concrete tank) for mechanically anchored tanks or the deflection caused by bending of the bottom of self-anchored tanks. The horizontal movements listed are caused by the deformation of the tank at the base.

In addition, interconnected equipment, walkways, and bridging between multiple tanks must be designed to resist the loads and accommodate the displacements imposed by seismic forces. Unless connected tanks and vessels are founded on a common rigid foundation, the calculated differential movements must be assumed to be out of phase.

C15.7.5 Anchorage. Many steel tanks can be designed without anchors by using annular plate detailing in accordance with reference documents. Where tanks must be anchored because of overturning potential, proper anchorage design provides both a shell attachment and an embedment detail that allows the bolt to yield without tearing the shell or pulling the bolt out of the foundation. Properly designed anchored tanks have greater reserve strength to resist seismic overload than do unanchored tanks.

To ensure that the bolt yields (stretches) before failure of the anchor embedment, the anchor embedment must be designed in accordance with ACI 318, Eq. (17.4.1.2), and must be provided with a minimum gauge length of eight bolt diameters. Gauge length is the length of the bolt that is allowed to stretch. It may include part of the embedment length into the concrete that is not bonded to the bolt. A representation of gauge length is shown in Fig. C15.7-1.

It is also important that the bolt not be significantly oversized to ensure that the bolt stretches. The prohibition on using the load combinations with overstrength of Section 12.4.3 is intended to accomplish this goal.

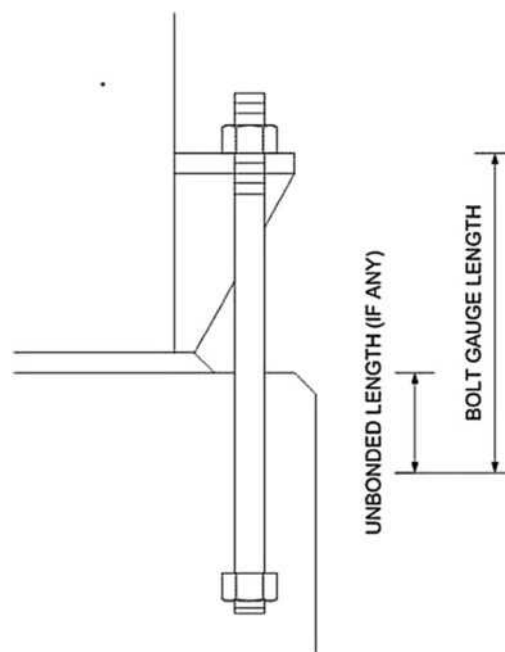


FIGURE C15.7-1 Bolt gauge length

Where anchor bolts and attachments are misaligned such that the anchor nut or washer does not bear evenly on the attachment, additional bending stresses in threaded areas may cause premature failure before anchor yielding.

C15.7.6 Ground-Supported Storage Tanks for Liquids

C15.7.6.1 General. The response of ground storage tanks to earthquakes is well documented by Housner (1963), Wozniak and Mitchell (1978), Velesztos (1974), and others. Unlike building structures, the structural response of these tanks is influenced strongly by the fluid–structure interaction. Fluid–structure interaction forces are categorized as sloshing (convective) and rigid (impulsive) forces. The proportion of these forces depends on the geometry (height-to-diameter ratio) of the tank. API 650, API 620, AWWA D100-11, AWWA D110, AWWA D115, and ACI 350.3 provide the data necessary to determine the relative masses and moments for each of these contributions.

The standard requires that these structures be designed in accordance with the prevailing reference documents, except that the height of the sloshing wave, δ_s , must be calculated using Eq. (15.7-13). API 650 and AWWA D100-11 include this requirement in their latest editions.

Eqs. (15.7-10) and (15.7-11) provide the spectral acceleration of the sloshing liquid for the constant-velocity and constant-displacement regions of the response spectrum, respectively. The 1.5 factor in these equations is an adjustment for 0.5% damping. An exception in the use of Eq. (15.7-11) was added for the 2010 edition of this standard. The mapped values of T_L were judged to be unnecessarily conservative by the ASCE 7 Seismic Subcommittee in light of actual site-specific studies carried out since the introduction of the T_L requirements of ASCE/SEI 7-05. These studies indicate that the mapped values of T_L appear to be very conservative based on observations during recent large earthquakes, especially the 2010 M_w 8.8 Chilean earthquake, where the large amplifications at very long periods (6–10 s) were not evident either in the ground motion records or in the behavior of long-period structures (particularly sloshing in tanks). Because a revision of the T_L maps is a time-consuming task that was not

possible during the 2010 update cycle, an exception was added to allow the use of site-specific values that are less than the mapped values with a floor of 4 s or one-half the mapped value of T_L . The exception was added under Section 15.7.6 because, for nonbuilding structures, the overly conservative values for T_L are primarily an issue for tanks and vessels. Discussion of the site-specific procedures can be found in the Commentary for Chapter 22.

Small-diameter tanks and vessels are more susceptible to overturning and vertical buckling. As a general rule, a greater ratio of H/D produces lower resistance to vertical buckling. Where H/D is greater than 2, overturning approaches “rigid mass” behavior (the sloshing mass is small). Large-diameter tanks may be governed by additional hydrodynamic hoop stresses in the middle regions of the shell.

The impulsive period (the natural period of the tank components and the impulsive component of the liquid) is typically in the 0.25–0.6 s range. Many methods are available for calculating the impulsive period. The Veletsos flexible-shell method is commonly used by many tank designers. For example, see Veletsos (1974) and Malhotra et al. (2000).

C15.7.6.1.1 Distribution of Hydrodynamic and Inertia Forces. Most of the reference documents for tanks define reaction loads at the base of the shell–foundation interface, without indicating the distribution of loads on the shell as a function of height. ACI 350.3 specifies the vertical and horizontal distribution of such loads.

The overturning moment at the base of the shell in the industry reference documents is only the portion of the moment that is transferred to the shell. The total overturning moment also includes the variation in bottom pressure, which is an important consideration for design of pile caps, slabs, or other support elements that must resist the total overturning moment. Wozniak and Mitchell (1978) and U.S. Department of Energy TID-7024 (1963) provide additional information.

C15.7.6.1.2 Sloshing. In past earthquakes, sloshing contents in ground storage tanks have caused both leakage and noncatastrophic damage to the roof and internal components. Even this limited damage and the associated costs and inconvenience can be significantly mitigated where the following items are considered:

1. Effective masses and hydrodynamic forces in the container;
2. Impulsive and pressure loads at
 - a. The sloshing zone (that is, the upper shell and edge of the roof system);
 - b. The internal supports (such as roof support columns and tray supports); and
 - c. The internal equipment (such as distribution rings, access tubes, pump wells, and risers); and
3. Freeboard (which depends on the sloshing wave height).

When no freeboard is required, a minimum freeboard of $0.7\delta_s$ is recommended for economic considerations. Freeboard is always required for tanks assigned to Risk Category IV.

Tanks and vessels storing biologically or environmentally benign materials typically do not require freeboard to protect the public health and safety. However, providing freeboard in areas of frequent seismic occurrence for vessels normally operated at or near top capacity may lessen damage (and the cost of subsequent repairs) to the roof and upper container. The exception to the minimum required freeboard per Table 15.7-3 for open-top tanks was added because it is rare for damage to occur that would impair the functionality of the facility when water or municipal wastewater overtops an open-top tank, provided that measures have been taken to intercept and properly handle the resulting overflow.

The sloshing liquid height specified in Section 15.7.6.1.2 is based on the design earthquake defined in the standard. For economic reasons, freeboard for tanks assigned to Risk Category I, II, or III may be calculated using a fixed value of T_L equal to 4 s (as indicated in Section 15.7.6.1.2, c) but using the appropriate Importance Factor taken from Table 1.5-2. Because of life-safety and operational functionality concerns, freeboard for tanks assigned to Risk Category IV must be based on the mapped value of T_L . Because use of the mapped value of T_L results in the theoretical maximum value of freeboard, the calculation of freeboard in the case of Risk Category IV tanks is based on an Importance Factor equal to 1.0 (as indicated in Section 15.7.6.1.2 b).

If the freeboard provided is less than the computed sloshing height, δ_s , the sloshing liquid impinges on the roof in the vicinity of the roof-to-wall joint, subjecting it to a hydrodynamic force. This force may be approximated by considering the sloshing wave as a hypothetical static liquid column that has a height, δ_s . The pressure exerted at any point along the roof at a distance y_s above the at-rest surface of the stored liquid may be assumed to be equal to the hydrostatic pressure exerted by the hypothetical liquid column at a distance $\delta_s - y_s$ from the top of that column. A better approximation of the pressure exerted on the roof is found in Malhotra (2005, 2006).

Another effect of a less-than-full freeboard is that the restricted convective (sloshing) mass “converts” into an impulsive mass, thus increasing the impulsive forces. This effect should be taken into account in the tank design. A method for converting the restricted convective mass into an impulsive mass is found in Malhotra (2005, 2006). It is recommended that sufficient freeboard to accommodate the full sloshing height be provided wherever possible.

Eq. (15.7-13) was revised to use the theoretical formulation for sloshing wave height instead of the rounded value introduced in ASCE/SEI 7-05. The rounded value of Eq. (15.6-1) increased the required freeboard by approximately 19%, thereby significantly increasing the cost of both secondary containment and large-diameter, ground-supported storage tanks. See Section C15.6.5.1 for additional commentary on freeboard.

C15.7.6.1.4 Internal Elements. Wozniak and Mitchell (1978) provide a recognized analysis method for determining the lateral loads on internal components caused by sloshing liquid.

C15.7.6.1.5 Sliding Resistance. Historically, steel ground-supported tanks full of product have not slid off foundations. A few unanchored, empty tanks or bulk storage tanks without steel bottoms have moved laterally during earthquake ground shaking. In most cases, these tanks may be returned to their proper locations. Resistance to sliding is obtained from the frictional resistance between the steel bottom and the sand cushion on which bottoms are placed. Because tank bottoms usually are crowned upward toward the tank center and are constructed of overlapping, fillet-welded, individual steel plates (resulting in a rough bottom), it is reasonably conservative to take the ultimate coefficient of friction on concrete as 0.70 (AISC 1986), and therefore a value of $\tan 30^\circ (= 0.577)$ for sand is used in design. The value of 30° represents the internal angle of friction of sand and is conservatively used in design. The vertical weight of the tank and contents, as reduced by the component of vertical acceleration, provides the net vertical load. An orthogonal combination of vertical and horizontal seismic forces, following the procedure in Section 12.5.3, may be used. In recent years, a significant issue has been the prevention of subsurface pollution caused by tank bottom corrosion and leakage. To prevent this problem, liners are often used with the tank

foundation. When some of these liners are used, sliding of the tank and/or foundation caused by the seismic base shear may be an issue. If the liner is completely contained within a concrete ring-wall foundation, the liner's surface is not the critical plane to check for sliding. If the liner is placed within an earthen foundation or is placed above or completely below a concrete foundation, it is imperative that sliding be evaluated. It is recommended that the sliding resistance factor of safety be at least 1.5.

C15.7.6.1.6 Local Shear Transfer. The transfer of seismic shear from the roof to the shell and from the shell to the base is accomplished by a combination of membrane shear and radial shear in the wall of the tank. For steel tanks, the radial (out-of-plane) seismic shear is very small and usually is neglected; thus, the shear is assumed to be resisted totally by membrane (in-plane) shear. For concrete walls and shells, which have a greater radial shear stiffness, the shear transfer may be shared. The ACI 350.3-06 (2006) commentary provides further discussion.

C15.7.6.1.7 Pressure Stability. Internal pressure may increase the critical buckling capacity of a shell. Provision to include pressure stability in determining the buckling resistance of the shell for overturning loads is included in AWWA D100-11 (2011). Recent testing on conical and cylindrical shells with internal pressure yielded a design methodology for resisting permanent loads in addition to temporary wind and seismic loads (Miller et al. 1997).

C15.7.6.1.8 Shell Support. Anchored steel tanks should be shimmed and grouted to provide proper support for the shell and to reduce impact on the anchor bolts under reversible loads. The high bearing pressures on the toe of the tank shell may cause inelastic deformations in compressible material (such as fiberboard), creating a gap between the anchor and the attachment. As the load reverses, the bolt is no longer snug and an impact of the attachment on the anchor can occur. Grout is a structural element and should be installed and inspected as an important part of the vertical and lateral force-resisting system.

C15.7.6.1.9 Repair, Alteration, or Reconstruction. During their service life, storage tanks are frequently repaired, modified, or relocated. Repairs often are related to corrosion, improper operation, or overload from wind or seismic events. Modifications are made for changes in service, updates to safety equipment for changing regulations, or installation of additional process piping connections. It is imperative that these repairs and modifications be designed and implemented properly to maintain the structural integrity of the tank or vessel for seismic loads and the design operating loads.

The petroleum steel tank industry has developed specific guidelines in API 653 that are statutory requirements in some states. It is recommended that the provisions of API 653 also be applied to other liquid storage tanks (e.g., water, wastewater, and chemical) as it relates to repairs, modifications, or relocation that affect the pressure boundary or lateral force-resisting system of the tank or vessel.

C15.7.7 Water Storage and Water Treatment Tanks and Vessels. The AWWA design requirements for ground-supported steel water storage structures use allowable stress design procedures that conform to the requirements of the standard.

C15.7.7.1 Welded Steel. AWWA D100 refers to ASCE 7-05 and repeats the ASCE 7-05 seismic design ground motion maps within the body of the document. A requirement is added in this section to point the user to the ground motions in the current version of ASCE 7. The clause in AWWA D100, Section 13.5.4.4, "unless otherwise specified" in the context of the determination of

seismic freeboard can result in seismic freeboard below that required by ASCE 7 and is therefore disallowed.

C15.7.7.2 Bolted Steel. A clarification on the ground motions to use in design is added and restrictions are added on the use of Type 6 tanks in AWWA D103 (2009). AWWA D103 refers to ASCE 7-05 and repeats the ASCE 7-05 ground motion maps within the body of the document. Therefore, a clarifying statement is added to point the user to the seismic design ground motions in the current version of ASCE 7. A Type 6 tank is a concrete-bottom bolted steel shell tank with an embedded steel base setting ring. Type 6 tanks are considered to be mechanically anchored. There are no requirements for the anchorage design or bottom design (other than ACI 318) in AWWA D103. For the tank to be considered mechanically anchored, the tank bottom cannot uplift. In this case, the tank bottom is the foundation. If the bottom/foundation uplifts, the tank is now a self-anchored tank and the additional shell compression that develops must be taken into account in the design. That is why J in equation 14-32 of AWWA D103 (2009) is limited to 0.785.

C15.7.7.3 Reinforced and Prestressed Concrete. A review of ACI 350.3 (2006), *Seismic Design of Liquid-Containing Concrete Structures and Commentary*, revealed that this document is not in general agreement with the seismic provisions of ASCE/SEI 7-10.

This section was clarified to note that the Importance Factor, I , and the response modification factor, R , are to be specified by ASCE/SEI 7 and not the reference document. The descriptions used in ACI 350.3 to determine the applicable values of the Importance Factor and response modification factor do not match those used in ASCE/SEI 7.

It was noted that the ground motions for determining the convective (sloshing) seismic forces specified in ACI 350.3 were not the same and are actually lower than those specified by ASCE/SEI 7. ACI 350.3 essentially redefines the long-period transition period, T_L . This alternate transition period allows large-diameter tanks to have significantly lower convective forces and lower seismic freeboard than those permitted by the provisions of ASCE/SEI 7. Therefore, Section 15.7.7.3 was revised to require that the convective acceleration be determined according to the procedure found in Section 15.7.6.1.

C15.7.8 Petrochemical and Industrial Tanks and Vessels Storing Liquids

C15.7.8.1 Welded Steel. The American Petroleum Institute (API) uses an allowable stress design procedure that conforms to the requirements of the standard.

The most common damage to tanks observed during past earthquakes includes the following:

1. Buckling of the tank shell near the base because of excessive axial membrane forces. This buckling damage is usually evident as "elephant foot" buckles a short distance above the base or as diamond-shaped buckles in the lower ring. Buckling of the upper ring also has been observed.
2. Damage to the roof caused by impingement on the underside of the roof of sloshing liquid with insufficient freeboard.
3. Failure of piping or other attachments that are overly restrained.
4. Foundation failures.

Other than the above damage, the seismic performance of floating roofs during earthquakes has generally been good, with damage usually confined to the rim seals, gauge poles, and ladders. However, floating roofs have sunk in some earthquakes

because of lack of adequate freeboard or the proper buoyancy and strength required by API 650. Similarly, the performance of open-top tanks with top wind girder stiffeners designed per API 650 has been generally good.

C15.7.8.2 Bolted Steel. Bolted steel tanks are often used for temporary functions. Where use is temporary, it may be acceptable to the jurisdictional authority to design bolted steel tanks for no seismic loads or for reduced seismic loads based on a reduced return period. For such reduced loads based on reduced exposure time, the owner should include a signed removal contract with the fixed removal date as part of the submittal to the authority having jurisdiction.

C15.7.9 Ground-Supported Storage Tanks for Granular Materials

C15.7.9.1 General. The response of a ground-supported storage tank storing granular materials to a seismic event is highly dependent on its height-to-diameter (H/D) ratio and the characteristics of the stored product. The effects of intergranular friction are described in more detail in Section C15.7.9.3.1 (increased lateral pressure), C15.7.9.3.2 (effective mass), and C15.7.9.3.3 (effective density).

Long-term increases in shell hoop tension because of temperature changes after the product has been compacted also must be included in the analysis of the shell; Anderson (1966) provides a suitable method.

C15.7.9.2 Lateral Force Determination. Seismic forces acting on ground-supported liquid storage tanks are divided between impulsive and convective (sloshing) components. However, in a ground-supported storage tank for granular materials, all seismic forces are of the impulsive type and relate to the period of the storage tank itself. Because of the relatively short period of a tank shell, the response is normally in the constant acceleration region of the response spectrum, which relates to S_{DS} . Therefore, the seismic base shear is calculated as follows:

$$V = \frac{S_{DS}}{\left(\frac{R}{I}\right)} W_{\text{effective}} \quad (\text{C15.7-3})$$

where V , S_{DS} , I , and R have been previously defined, and $W_{\text{effective}}$ is the gross weight of the stored product multiplied by an effective mass factor and an effective density factor, as described in Sections C15.7.9.3.2 and C15.7.9.3.3, plus the dead weight of the tank. Unless substantiated by testing, it is recommended that the product of the effective mass factor and the effective density factor be taken as no less than 0.5 because of the limited test data and the highly variable properties of the stored product.

C15.7.9.3 Force Distribution to Shell and Foundation

C15.7.9.3.1 Increased Lateral Pressure. In a ground-supported tank storing granular materials, increased lateral pressures develop as a result of rigid body forces that are proportional to ground acceleration. Information concerning design for such pressure is scarce. Trahair et al. (1983) describe both a simple, conservative method and a difficult, analytical method using failure wedges based on the Mononobe–Okabe modifications of the classical Coulomb method.

C15.7.9.3.2 Effective Mass. For ground-supported tanks storing granular materials, much of the lateral seismic load can be transferred directly into the foundation, via intergranular shear, before it can reach the tank shell. The effective mass that loads

the tank shell is highly dependent on the H/D ratio of the tank and the characteristics of the stored product. Quantitative information concerning this effect is scarce, but Trahair et al. (1983) describe a simple, conservative method to determine the effective mass. That method presents reductions in effective mass, which may be significant, for H/D ratios less than 2. This effect is absent for elevated tanks.

C15.7.9.3.3 Effective Density. Granular material stored in tanks (both ground-supported and elevated) does not behave as a solid mass. Energy loss through intergranular movement and grain-to-grain friction in the stored material effectively reduces the mass subject to horizontal acceleration. This effect may be quantified by an effective density factor less than 1.0.

Based on Chandrasekaran and Jain (1968) and on shake table tests reported in Chandrasekaran et al. (1968), ACI 313 (1997) recommends an effective density factor of not less than 0.8 for most granular materials. According to Chandrasekaran and Jain (1968), an effective density factor of 0.9 is more appropriate for materials with high moduli of elasticity, such as aggregates and metal ores.

C15.7.9.3.4 Lateral Sliding. Most ground-supported steel storage tanks for granular materials rest on a base ring and do not have a steel bottom. To resist seismic base shear, a partial bottom or annular plate is used in combination with anchor bolts or a curb angle. An annular plate can be used alone to resist the seismic base shear through friction between the plate and the foundation, in which case the friction limits of Section 15.7.6.1.5 apply. The curb angle detail serves to keep the base of the shell round while allowing it to move and flex under seismic load. Various base details are shown in Fig. 13 of Kaups and Lieb (1985).

C15.7.9.3.5 Combined Anchorage Systems. This section is intended to apply to combined anchorage systems that share loads based on their relative stiffnesses, and not to systems where sliding is resisted completely by one system (such as a steel annular plate) and overturning is resisted completely by another system (such as anchor bolts).

C15.7.10 Elevated Tanks and Vessels for Liquids and Granular Materials

C15.7.10.1 General. The three basic lateral load-resisting systems for elevated water tanks are defined by their support structure:

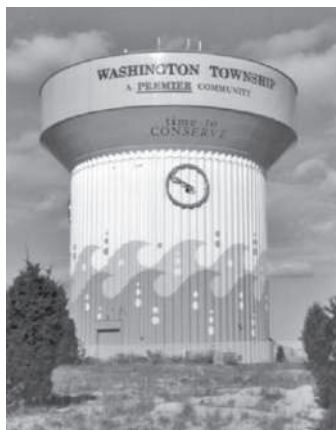
1. Multilegged braced steel tanks (trussed towers, as shown in Fig. C15.1-1);
2. Small-diameter, single-pedestal steel tank (cantilever column, as shown in Fig. C15.7-2); and
3. Large-diameter, single-pedestal tanks of steel or concrete construction (load-bearing shear walls, as shown in Fig. C15.7-3).

Unbraced multilegged tanks are uncommon. These types of tanks differ in their behavior, redundancy, and resistance to overload. Multilegged and small-diameter pedestal tanks have longer fundamental periods (typically greater than 2 s) than the shear wall type tanks (typically less than 2 s). The lateral load failure mechanisms usually are brace failure for multilegged tanks, compression buckling for small-diameter steel tanks, compression or shear buckling for large-diameter steel tanks, and shear failure for large-diameter concrete tanks. Connection, welding, and reinforcement details require careful attention to mobilize the full strength of these structures. To provide a greater margin of safety, R factors used with elevated tanks typically are less than those for other comparable lateral load-resisting systems.



FIGURE C15.7-2 Small-diameter, single-pedestal steel tank

Source: Courtesy of CB&I LLC; reproduced with permission.



(a) Steel



(b) Concrete

FIGURE C15.7-3 Large-diameter, single-pedestal tank

Source: Courtesy of CB&I LLC; reproduced with permission.

C15.7.10.4 Transfer of Lateral Forces into Support Tower. The vertical loads and shears transferred at the base of a tank or vessel supported by grillage or beams typically vary around the base because of the relative stiffness of the supports, settlements, and variations in construction. Such variations must be considered in the design for vertical and horizontal loads.

C15.7.10.5 Evaluation of Structures Sensitive to Buckling Failure. Nonbuilding structures that are designed with limited structural redundancy for lateral loads may be susceptible to total failure when loaded beyond the design loads. This phenomenon is particularly true for shell-type structures that exhibit unstable postbuckling behavior, such as tanks and vessels supported on shell skirts or pedestals. Evaluation for this critical condition ensures stability of the nonbuilding structure for governing design loads.

The design spectral response acceleration, S_a , used in this evaluation includes site factors. The I/R coefficient is taken as 1.0 for this critical check. The structural capacity of the shell is

taken as the critical buckling strength (that is, the factor of safety is 1.0). Vertical and orthogonal combinations need not be considered for this evaluation because the probability of peak values occurring simultaneously is very low.

The intent of Section 15.7.10.5 and Table 15.4-2 is that skirt-supported vessels must be checked for seismic loads based on $I_e/R = 1.0$ if the structure falls in Risk Category IV or if an R value of 3.0 is used in the design of the vessel. For the purposes of this section, a skirt is a thin-walled steel cylinder or cone used to support the vessel in compression. Skirt-supported vessels fail in buckling, which is not a ductile failure mode. Therefore, a more conservative design approach is required. The $I_e/R = 1.0$ check typically governs the design of the skirt over using loads determined with an R factor of 3 in a moderate to high area of seismic activity. The only benefit of using an R factor of 3 in this case is in the design of the foundation. The foundation is not required to be designed for the $I_e/R = 1.0$ load. Section 15.7.10.5, item b, states that resistance of the structure shall be defined as the critical buckling resistance of the element for the $I_e/R = 1.0$ load. This stipulation means that the support skirt can be designed based on critical buckling (factor of safety of 1.0). The critical buckling strength of a skirt can be determined using a number of published sources. The two most common methods for determining the critical buckling strength of a skirt are the ASME BVPC (2007), Section VIII, Division 2, 2008 Addenda, Paragraph 4.4, using a factor of safety of 1.0 and AWWA D100-05 (2006a), Section 13.4.3.4. To use these methods, the radius, length, and thickness of the skirt; modulus of elasticity of the steel; and yield strength of the steel are required. These methods take into account both local buckling and slenderness effects of the skirt. Under no circumstance should the theoretical buckling strength of a cylinder, found in many engineering mechanics texts, be used to determine the critical buckling strength of the skirt. The theoretical value, based on a perfect cylinder, does not take into account imperfections built into real skirts. The theoretical buckling value is several times greater than the actual value measured in tests. The buckling values found in the suggested references above are based on actual tests.

Examples of applying the ASME BVPC (2007), Section VIII, Division 2, 2008 Addenda, Paragraph 4.4, and AWWA D100-05 (2006a), Section 13.4.3.4, buckling rules are shown in Figs. C15.7-4 and C15.7-5.

C15.7.10.7 Concrete Pedestal (Composite) Tanks. A composite elevated water storage tank is composed of a welded steel tank for watertight containment, a single-pedestal concrete support structure, a foundation, and accessories. The lateral load-resisting system is a load-bearing concrete shear wall. ACI 371R (1998), referenced in previous editions of ASCE 7, has been replaced with AWWA D107 (2010). Because AWWA D107-10 is based on the seismic design ground motions from ASCE 7-05, a requirement was added in Section 15.7.10.7 to require the use of the seismic design ground motions from Section 11.4.

C15.7.11 Boilers and Pressure Vessels. The support system for boilers and pressure vessels must be designed for the seismic forces and displacements presented in the standard. Such design must include consideration of the support, the attachment of the support to the vessel (even if “integral”), and the body of the vessel itself, which is subject to local stresses imposed by the support connection.

C15.7.12 Liquid and Gas Spheres. The commentary in Section C15.7.11 also applies to liquid and gas spheres.

C15.7.13 Refrigerated Gas Liquid Storage Tanks and Vessels. Even though some refrigerated storage tanks and

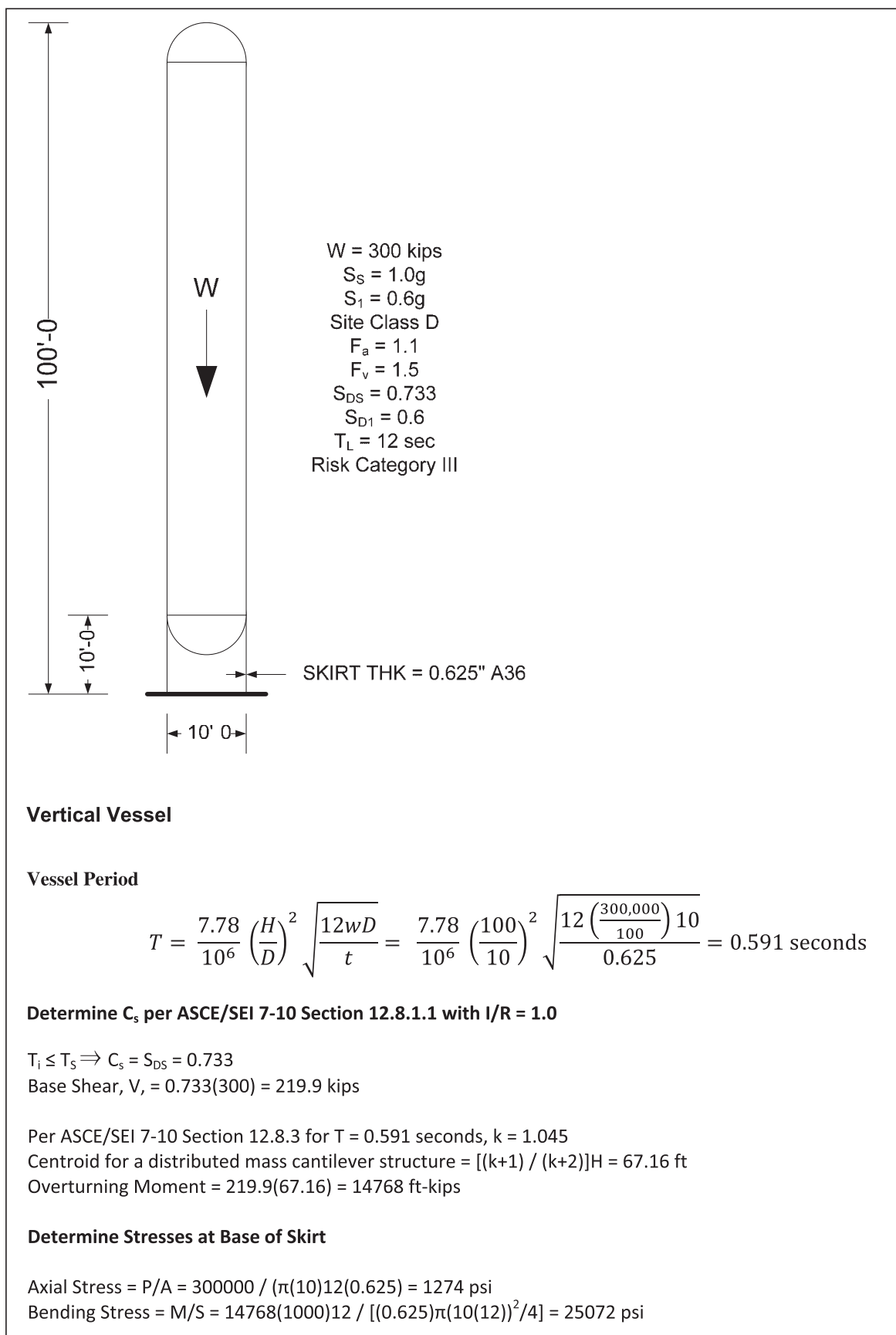


FIGURE C15.7-4 Example Problem Using ASME BPVC (2007), Section VIII, Division 2, 2008 Addenda, Paragraph 4.4

continues

Input Data for ASME Section VIII, Div. 2 Buckling Checks (Paragraph 4.4)

Input Values

Course =	Skirt
t = thickness of vessel section =	0.625 in.
H _T = top elevation of course =	120 in.
H _B = bottom elevation of course =	0 in.
D _O = outer diameter of vessel section =	120 in.
E _y = material modulus of elasticity =	29,000,000 psi
S _y = material yield strength =	36,000 psi
P _{ext} = external pressure =	0.000 psi
f _a = axial comp membrane stress from axial load =	1,274 psi
f _b = axial comp membrane stress from bending =	25,072 psi
V = net section shear force =	219,900 lbs
V _{phi} = applied shear force angle =	90 deg.
C _m = coefficient =	1 0.85, 1, or 0.6 – 0.4(M ₁ /M ₂)
K _u = effective length factor =	2.1 Free-Fixed
L _u = maximum unbraced length =	1,200 in.
L = design length vessel section for external pressure =	120 in.
L = design length vessel section for axial compression =	120 in.
FS = Input Factor of Safety =	1.0

Calculated Values

R _o = outer radius of shell section =	60 in.
R = radius to center of shell =	59.6875 in.
R _m = vessel mean radius =	59.6875 in.
r = radius of gyration of cyl = $(1/4)(D_o^2 + D_i^2)^{0.5}$ =	42.2 in.
A = Cross sectional area of cylinder =	234.4 in ²
f _q = axial compressive membrane stress = $P_{ext}\pi D_i^2/4A$ =	0.0 psi

4.4.12 Combined Loadings and Allowable Compressive Stresses (continued)

- b) Axial Compressive Stress Acting Alone - The allowable axial compressive membrane stress of a cylinder subject to an axial compressive load acting alone, F_{xa} , is computed by following equations.

1) For $\lambda_c \leq 0.15$ (Local Buckling)

$$F_{xa} = \min [F_{xa1}, F_{xa2}] \quad (4.4.61)$$

$$F_{xa1} = S_y / FS \quad \text{for } D_o/t \leq 135 \quad (4.4.62)$$

$$F_{xa1} = 466 S_y / [(331 + D_o/t) FS] \quad \text{for } 135 < D_o/t < 600 \quad (4.4.63)$$

$$F_{xa1} = 0.5 S_y / FS \quad \text{for } 135 < D_o/t < 600 \quad (4.4.64)$$

$$F_{xa2} = F_{xe} / FS \quad (4.4.65)$$

$$\text{where: } F_{xe} = C_x E_y t / D_o \quad (4.4.66)$$

$$C_x = \min[409 c / [389 + D_o/t], 0.9] \quad \text{for } D_o/t < 1247 \quad (4.4.67)$$

$$C_x = 0.25 c \quad \text{for } 1247 \leq D_o/t \leq 2000 \quad (4.4.68)$$

$$c = 2.64 \quad \text{for } M_x \leq 1.5 \quad (4.4.69)$$

$$c = 3.13 / M_x^{0.42} \quad \text{for } 1.5 < M_x < 15 \quad (4.4.70)$$

$$c = 1.0 \quad \text{for } M_x \geq 15 \quad (4.4.71)$$

FIGURE C15.7-4 (Continued) Example Problem Using ASME BPVC (2007), Section VIII, Division 2, 2008 Addenda, Paragraph 4.4

continues

4.4.12 Combined Loadings and Allowable Compressive Stresses (continued)

$$M_x = L/(R_0 t)^{1/2} \quad (4.4.124)$$

where L is the design length of a vessel section between lines of support

$$D_o/t = 192.00 \quad 135 < D_o/t < 600$$

$$M_x = L/(R_0 t)^{1/2} = 19.60 > 15 \quad c = 1.0000$$

$$D_o/t < 1247 \quad C_x = \min [409 c / [389 + D_o/t], 0.9] = 0.7039587 \quad (4.4.67)$$

$$F_{xe} = C_x E_y t / D_o = 106,327 \text{ psi} \quad (4.4.66)$$

Calculate F_{xa1}

$$F_{ic} = 466 S_y / (331 + D_o/t) = 32,076 \text{ psi}$$

Use Input FS = 1.0

$$F_{xa1} = 466 S_y / [(331 + D_o/t) FS] = 32,076 \text{ psi}$$

Calculate F_{xa2}

$$F_{ic} = 106,327 \text{ psi}$$

Use Input FS = 1.0

$$F_{xa2} = F_{xe} / FS = 106,327 \text{ psi} \quad (4.4.65)$$

Calculate F_{xa}

$$F_{xa} = \min [F_{xa1}, F_{xa2}] = 32,076 \text{ psi} \quad (4.4.61)$$

$$\lambda_{dc} = (K) (L_u) / [(\pi)(r_g)] [(F_{xa} / FS) / E]^{0.5} = 0.6321 \quad 0.15 < \lambda_{dc} < 1.147$$

2) For $\lambda_{dc} > 0.15$ and $K_u L_u / r_g < 200$ (Column Buckling)

$$F_{ca} = F_{xa} (1 - 0.74 (\lambda_{dc} - 0.15))^{0.3} \quad 0.15 < \lambda_{dc} < 1.147 \quad (4.4.72)$$

$$F_{ca} = 0.88 F_{xa} / (\lambda_{dc})^2 \quad \lambda_{dc} \geq 1.147 \quad (4.4.73)$$

$$K L_u / r_g = 59.7 < 200$$

$\lambda_{dc} > 0.15$ and $K L_u / r_g < 200$ therefore:

$$F_{ca} = F_{xa} (1 - 0.74 (\lambda_{dc} - 0.15))^{0.3} = 28,100 \text{ psi}$$

- c) Compressive Bending Stress - The allowable axial compressive membrane stress of a cylindrical shell subject to a bending moment acting across the full circular cross section F_{ba} , is computed using the following equations.

$$F_{ba} = F_{xa} \quad \text{for } 135 \leq D_o / t \leq 2000 \quad (4.4.74)$$

$$F_{ba} = 466 S_y / [(331 + D_o / t) FS] \quad \text{for } 100 < D_o / t < 135 \quad (4.4.75)$$

$$F_{ba} = 1.081 S_y / FS \quad \text{for } D_o / t < 100 \text{ and } y > 0.11 \quad (4.4.76)$$

$$F_{ba} = (1.4 - 2.9 y) S_y / FS \quad \text{for } D_o / t < 100 \text{ and } y < 0.11 \quad (4.4.77)$$

$$\text{where: } y = S_y D_o / E_y t \quad (4.4.78)$$

$$D_o / t = 192$$

$$y = S_y D_o / E_y t = 0.2383$$

$$F_{ic} = F_{xa}$$

$$F_{ic} = 32,076 \text{ psi}$$

$$D_o / t = 192 > 135 \text{ (see Sect. 3.1.1)}$$

$$\text{Use Input FS} = 1.0$$

$$F_{ba} = F_{xa} = 32,076 \text{ psi}$$

FIGURE C15.7-4 (Continued) Example Problem Using ASME BPVC (2007), Section VIII, Division 2, 2008 Addenda, Paragraph 4.4

continues

- d) Shear Stress - The allowable shear stress of a cylindrical shell, F_{va} , is computed using the following equations.

$$F_{va} = n_v F_{ve} / FS \quad (4.4.79)$$

$$\text{where: } F_{ve} = a_v C_v E t / D_o \quad (4.4.80)$$

$$C_v = 4.454 \quad \text{for } M_x < 1.5 \quad (4.4.81)$$

$$C_v = (9.64 / M_x^2) (1 + 0.0239 M_x^3)^{1/2} \quad \text{for } 1.5 < M_x < 26 \quad (4.4.82)$$

$$C_v = 1.492 / (M_x)^{1/2} \quad \text{for } 26 < M_x < 4.347 D_o / t \quad (4.4.83)$$

$$C_v = 0.716 (t / D_o)^{1/2} \quad \text{for } M_x > 4.347 D_o / t \quad (4.4.84)$$

$$a_v = 0.8 \quad \text{for } D_o / t < 500 \quad (4.4.85)$$

$$a_v = 1.389 - 0.218 \log_{10} (D_o / t) \quad \text{for } D_o / t > 500 \quad (4.4.86)$$

$$n_v = 1.0 \quad \text{for } F_{ve} / S_y < 0.48 \quad (4.4.87)$$

$$n_v = 0.43 S_y / F_{ve} + 0.1 \quad \text{for } 0.48 < F_{ve} / S_y < 1.7 \quad (4.4.88)$$

$$n_v = 0.6 S_y / F_{ve} \quad \text{for } F_{ve} / S_y > 1.7 \quad (4.4.89)$$

$$D_o / t = 192$$

$$M_x = L / (R_o t)^{0.5} = 19.596$$

$$1.5 < M_x < 26 \quad C_v = (9.64 / M_x^2) (1 + 0.0239 M_x^3)^{0.5} = 0.3376$$

$$D_o / t < 500 \quad a_v = 0.8000$$

$$F_{ve} = a_v C_v E t / D_o = 40,793 \text{ psi}$$

$$F_{ve} / S_y = 1.13 \quad 0.48 < F_{ve} / S_y < 1.7$$

$$n_v = 0.43 S_y / F_{ve} + 0.1 = 0.4795$$

$$F_{ic} = 19,559 \text{ psi}$$

$$\text{Use Input } FS = 1.000$$

$$F_{va} = n_v F_{ve} / FS = 19,559 \text{ psi}$$

4.4.12 Combined Loadings and Allowable Compressive Stresses (continued)

Axial Compressive Stress, Compressive Bending Stress, and Shear - the allowable compressive stress for the combination of uniform axial compression, axial compression due to bending, and shear in the absence of hoop compression.

$$\text{Let } K_s = 1 - (f_v / F_{va})^2 \quad (4.4.105)$$

$$\text{For } 0.15 < \lambda_{dc} < 1.2$$

$$\lambda_{dc} = 0.63 \quad (\text{Section 3.2}) \quad 0.15 < \lambda_{dc} < 1.2 \quad \text{OK}$$

$$f_a / (K_s F_{ca}) + (8 / 9) (\delta) f_b / (K_s F_{ba}) < 1.0 \quad f_a / (K_s F_{ca}) > 0.2 \quad (4.4.112)$$

$$f_a / (2 K_s F_{ca}) + (\delta) f_b / (K_s F_{ba}) < 1.0 \quad f_a / (K_s F_{ca}) < 0.2 \quad (4.4.113)$$

$$K_s = 1 - (f_v / F_{va})^2 = 0.9977$$

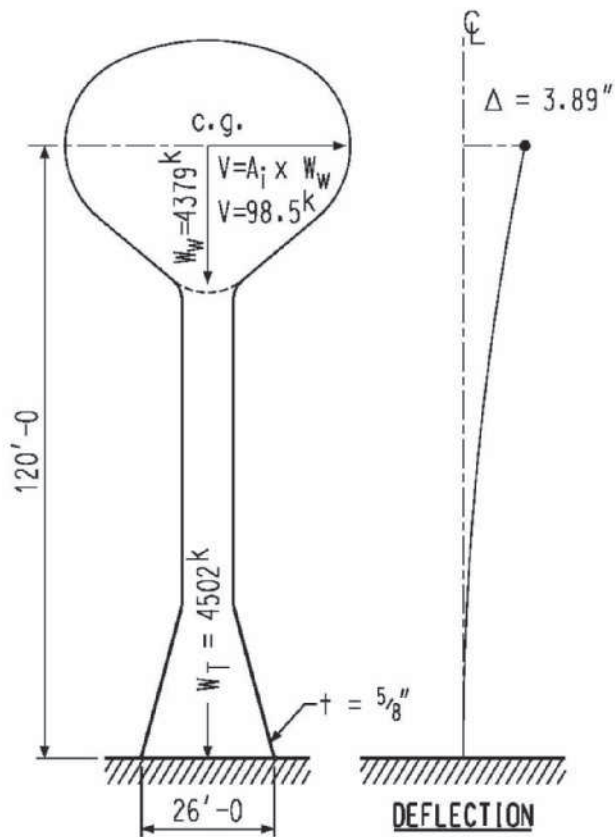
$$F_e = (\pi)^2 E / [K_s L_u / r]^2 = 80,287 \text{ psi}$$

$$\delta = C_m / [1 - f_a FS / F_e] = 1.0161$$

$$f_a / (K_s F_{ca}) = 0.045442959 < 0.2$$

$$f_a / (2 K_s F_{ca}) + (\delta) f_b / (K_s F_{ba}) = 0.82 < 1.0 \quad \text{OK!}$$

FIGURE C15.7-4 (Continued) Example Problem Using ASME BPVC (2007), Section VIII, Division 2, 2008 Addenda, Paragraph 4.4



Single Pedestal Elevated Water Tank

Seismic Information

$$S_s = 0.162, S_1 = 0.077$$

$$\text{Site Class C, } F_a = 1.2, F_v = 1.7$$

$$S_{DS} = 0.130, S_{D1} = 0.087, T_L = 12 \text{ s}$$

Risk Category IV

$$T_s = 0.674 \text{ s}$$

Tank Information

$$\text{Structure Period } T_i = 3.88 \text{ s}$$

$$\text{Class 2 Material: A36 } (F_y = 36 \text{ kip/in.}^2)$$

$$\text{Skirt angle (from vertical)} = 15 \text{ deg}$$

$$\text{Weight of tank and water, } W_w = 4,379 \text{ kip}$$

$$\text{Weight of tank, tower, and water, } W_t = 4,502 \text{ kip}$$

$$KL/r = 50$$

Determine S_{ai} per AWWA D100-05, Section 13.2.7.2

$$T_s < T_i \leq T_L \Rightarrow S_{ai} = S_{D1}/T_i = 0.087/3.88 = 0.0225$$

FIGURE C15.7-5 Example Problem Using AWWA D100-05, Section 13.4.3.4

continues

Determine Critical Buckling Acceleration ($I_p/1.4 R_i = 1$)

Per Section 13.4.3.4, $A_i = S_{ai}$ for critical buckling check (A_i in AWWA D100-05 is the same as C_s in ASCE/SEI 7-10)

$$A_i = 0.0225$$

Lateral Displacement Caused by S_{ai} ($P-\Delta$)

The final deflected position of the water centroid is an iterative process and must account for the additional moment applied to the structure because of the $P-\Delta$ effect. The deflection from the critical buckling deflection is equal to 3.89 in.

Check Skirt at Base of Tower

Seismic overturning moment at base of tower without $P-\Delta = 11,928$ ft-kip (includes mass of tower).

Seismic overturning moment at base of tower with $P-\Delta = 11,928$ ft-kip + $4,379$ kip $\times$ 3.89 in./ 12 in. per ft
 $= 13,348$ ft-kip

$$\text{Area of skirt} = \pi(26 \times 12)(0.625) = 612.6 \text{ in.}^2$$

$$\text{Section modulus of skirt} = \pi(26 \times 12)^2/4 \times 0.625 = 47,784 \text{ in.}^3$$

$$\text{Skirt stress caused by axial load} = 4,502(1,000)/(612.6 \times \cos 15) = 7,608 \text{ lb/in.}^2$$

$$\text{Skirt stress caused by moment} = 13,348(12)(1000)/(47,784 \times \cos 15) = 3,470 \text{ lb/in.}^2$$

Determine Critical Buckling Stress

$$R = 13 \times 12/\cos 15 = 161.5 \text{ in.}$$

$$t/R = 0.625/161.5 = 0.0039$$

For Class 2 material, $KL/r = 50$, and $t/R = 0.0039$, determine allowable axial compressive stress, F_a , from Table 13 of AWWA D100-05.

$$F_a = 9,882 \text{ lb/in.}^2$$

Per AWWA D100-05, Section 13.4.3.4,

$$\text{Critical buckling stress} = 2F_a = 19,764 \text{ lb/in.}^2$$

For Class 2 material and $t/R = 0.0039$, determine allowable bending compressive stress, F_b , from Table 11 of AWWA D100-05.

$$F_b = F_L = 10,380 \text{ lb/in.}^2$$

Per AWWA D100-05, Section 13.4.3.4,

$$\text{Critical bending stress} = 2F_b = 20,760 \text{ lb/in.}^2$$

Check unity per AWWA D100-05, Section 3.3.1:

$$7,608/19,764 + 3,470/20,760 = 0.552 \leq 1.0 \text{ OK}$$

FIGURE C15.7-5 (Continued) Example Problem Using AWWA D100-05, Section 13.4.3.4

vessels, such as those storing liquefied natural gas, are required to be designed for ground motions and performance goals in excess of those found in the standard, all such structures must also meet the requirements of this standard as a minimum. All welded steel refrigerated storage tanks and vessels must be designed in accordance with the requirements of the standard and the requirements of API 620.

C15.7.14 Horizontal, Saddle-Supported Vessels for Liquid or Vapor Storage. Past practice has been to assume that a horizontal, saddle-supported vessel (including its contents) behaves as a rigid structure (with natural period, T , less than 0.06 s). For this situation, seismic forces would be determined using the requirements of Section 15.4.2. For large horizontal, saddle-supported vessels (length-to-diameter ratio of 6 or more), this assumption can be unconservative, so Section 15.7.14.3 requires that the natural period be determined assuming the vessel to be a simply supported beam.

C15.8 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

Chapter 15 of this standard makes extensive use of reference documents in the design of nonbuilding structures for seismic forces; see Chapter 23. The documents referenced in Chapter 15 are industry documents commonly used to design specific types of nonbuilding structures. The vast majority of these reference documents contain seismic provisions that are based on the seismic ground motions of the 1997 Uniform Building Code (ICBO 1997) or earlier editions of the UBC. To use these reference documents, Chapter 15 modifies the seismic force provisions of these reference documents through the use of “bridging equations.” The standard only modifies industry documents that specify seismic demand and capacity. The bridging

equations are intended to be used directly with the other provisions of the specific reference documents. Unlike the other provisions of the standard, if the reference documents are written in terms of allowable stress design, then the bridging equations are shown in allowable stress design format. In addition, the detailing requirements referenced in Tables 15.4-1 and Table 15.4-2 are expected to be followed, as well as the general requirements found in Section 15.4.1. The usage of reference documents in conjunction with the requirements of Section 15.4.1 are summarized in Table C15.8-1.

Currently, only four reference documents have been revised to meet the seismic requirements of the standard. AWWA D100, API 620, API 650, and ANSI/RMI MH 16.1 have been adopted by reference in the standard without modification, except that height limits are imposed on “elevated tanks on symmetrically braced legs (not similar to buildings)” in AWWA D100, and the anchorage requirements of Section 15.4.9 are imposed on steel storage racks in ANSI/RMI MH 16.1. Three of these reference documents apply to welded steel liquid storage tanks.

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Table C15.8-1 Usage of Reference Documents in Conjunction with Section 15.4.1

Subject	Requirement
R , Ω_0 , and C_d values, detailing requirements, and height limits	Use values and limits in Tables 12.2-1, 15.4-1, or 15.4-2 as appropriate. Values from the reference document are not to be used.
Minimum base shear	Use the appropriate value from Eq. (15.4-1) or (15.4-2) for nonbuilding structures not similar to buildings. For structures containing liquids, gases, and granular solids supported at the base, the minimum seismic force cannot be less than that required by the reference document.
Importance Factor	Use the value from Section 15.4.1.1 based on Risk Category. Importance Factors from the reference document are not to be used unless they are greater than those provided in the standard.
Vertical distribution of lateral load	Use requirements of Section 12.8.3 or Section 12.9 or the applicable reference document.
Seismic provisions of reference documents	The seismic force provisions of reference documents may be used only if they have the same basis as Section 11.4 and the resulting values for total lateral force and total overturning moment are no less than 80% of the values obtained from the standard.
Load combinations	Load combinations specified in Section 2.3 (LRFD) or Section 15 (includes ASD load combinations of Section 2.4) must be used.

"FEMA 460. Building Seismic Safety Council, National Institute of Building Sciences, Washington, DC.

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API. (2014b). "Welded steel tanks for oil storage, 12th Ed., Addendum 1," *API 650*, Washington, DC.

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